DRÄXLMAIER

Interface Description

DRÄXLMAIER Shopfloor Systems

OC234-000019

OPC UA Template for CS & HV projects

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2 Version History

Version	Date	Modifications	Editor
3.00	22.11.2022	• New version of the document, based on “Interface Description - OPC UA Template for CS & HV projects 2.13”. • The OPC UA section “ProcessTimes” is now mandatory. • Renamed the OPC UA variable FunctionalGroup to FunctionalGroups. • Renamed the OPC UA variable MachineNumber to EquipmentMainNumber. • Renamed the OPC UA variable EquipmentNumber to EquipmentExemplarNumber. • Added the chapter “SecurityData”. • Added the OPC UA section “SecurityData” in the workplaces WORKPLACE_UNLOAD.“ • Removed deprecated energy related OPC UA variables (e.g. PowerCons, AirCons). Instead use the section “Energy”. • Added the OPC UA variable MES_JobNr in MES_ProductParameter section. • Concretize datatype for Date_And_Time (Siemens LDT). • Removed all references to the “Mandatory minimum set of process parameters”. • OPC UA section “MaterialConsumption”: Extended field descriptions for SAP Added parameter PositionIndex Added parameter SinglepartLabelSupplier	Graf Johanna IA252 Peter Keintzel OC234 Raab Thomas IA252
3.01	14.03.2023	• A reference to the term "IPST-Parameterliste" has been added. • Changed INTERRUPT logic from WORKPLACE_LOAD to WORKPLACE_UNLOAD • Added new fields MES_BookingError and MES_BookingErrorText to WORKPLACE_UNLOD and WORKPLACE_LOAD_UNLOAD section	Peter Keintzel OC234 Graf Johanna IA252
3.02	17.05.2023	• Added the OPC UA foundation member logo. • Extended document ○ “5.4.2.2 Status”: Added part counter values (see also chapter 4.3.9 in document InterfaceSpecification&Structure_OPC_UA_template_V3_03ff.) ○ “5.4.2.5.2 ProcessTimes”: Added Examples ○ Changed screenshots of OPC structures to match latest changes • Added chapters ○ “8 Initializing process- and traceability data”. ○ “9 Format of an error code for NOK and SCRAP components” containing error code definitions for EOL/HV/Runin workplaces • Changes on OPC Structure ○ Changed/added several predefined StringLengths ○ Renamed Real to Float ○ Added part counters in “5.4.2.2 Status “section ○ Added description to ActiveEnergy & ActivePower	Peter Keintzel OC234 Graf Johanna IA252

3.03	04.03.2024	• The topic “Multiple WORKPLACE_UNLOAD_XX sections” has been added. Several changes have been made throughout the document for this purpose. • Chapter “Alerts”: The name of a functional group has to be unique. Inactive alerts must be marked with AlertCode = -1, AlertLevel = -1 and AlertMessage = “”. • Added a hint for reordering the part to Interrupt process in “6.2.5.2 Workflow “Interrupt Process”“ • Changes on OPC Structure o Added _Count Variables for Alerts and MaterialConsumption in order to be able to reduce the data amount if needed o Renamed variable “Name” in FunctionalGroupsSection to “FunctionalGroupName” o Added field “AlternativeID” in GeneralData section, to store additional barcodes that can identify a part later on	Peter Keintzel OC234 Graf Johanna IA252
3.04	01.10.2024	• Updated image of a typical assembly line with several automatic cells in chapter “6.1 Introduction” • Optimize descriptions for chapter “6.2.5.1 Workflow “Provide Result Data“” • Added chapter “9 Reporting of process data for semifinished parts” • Added forbidden values in chapter “10 Format of an error code for NOK and SCRAP components” • Removed chapter “7.1 Malfunction of the MES Communication Interface” • Added chapter “6.2.4.1 WORKPLACE_CHECK” • Changed OPC Structure o Adopted various descriptions o Updated SecurityData section with new fields	Peter Keintzel OC234 Graf Johanna IA252

Table 1: Version History

3 Introduction

This OPC UA template specification contains the description of the communication protocol between an DRÄXLMAIER CS & HV assembly line and the DRÄXLMAIER Shopfloor Systems.

The communication is based on the platform independent OPC Unified Architecture (OPC UA) technology and uses OPC UA Data Access to control the production process.

The main tasks are:

- Serving the assembly line with all required data of an order.
- Enabling the assembly line to report the process results back to the DRÄXLMAIER Shopfloor Systems.
- Enabling the assembly line to report the current machine state like idle or running to the DRÄXLMAIER Shopfloor Systems.

Note

- This OPC UA template for CS & HV projects is a specific variant of the “DRÄXLMAIER Group MES - OPC UA Template”.
- It is based on the “DRÄXLMAIER Group MES - OPC UA Template” and is fully compatible.
- This document is a template. A more detailed project specific interface description will be developed during the project phase.
- Also see the documents in the section “Applicable documents” for further information and more details about the base specification “DRÄXLMAIER Group MES - OPC UA Template”.
- For a better understanding of this document, reading document “DRÄXLMAIER Group MES - OPC UA Template” is highly recommended.

4 Applicable Documents

Nr.	Name	Version	Description
1.	DRÄXLMAIER Group MES - OPC UA Template InterfaceSpecification&Structure_OPC_UA_template_V3_xx.pdf	3.05	The underlying base protocol. Contains technical details about the used OPC UA protocol and more detailed information about workflows.

Table 2: Applicable Documents

5 Information Model

The information model described in this document defines the required nodes and the hierarchically structure on the OPC UA server. It's also called "OPC UA Variable Tree".

5.1 Overview

The following picture shows an example of a typical machine interface based on the DRÄXLMAIER OPC UA information model for CS & HV projects:

- A global data section for reporting energy values, alerts, and PLC system information.
- A WORKPLACE_LOAD for receiving the order data, the material list, and additional parameters from MES.
- One or more WORKPLACE_XX for reporting the machine state of a single process.
- One or more WORKPLACE_YY for handling special requests between MES and machine within the normal production steps (e.g. validity checks (usually named WORKPLACE_CHECK), password requests (usually named WORKPLACE_GETPASSWORD),…).
- One or more WORKPLACE_UNLOAD sections for reporting process results back to MES.

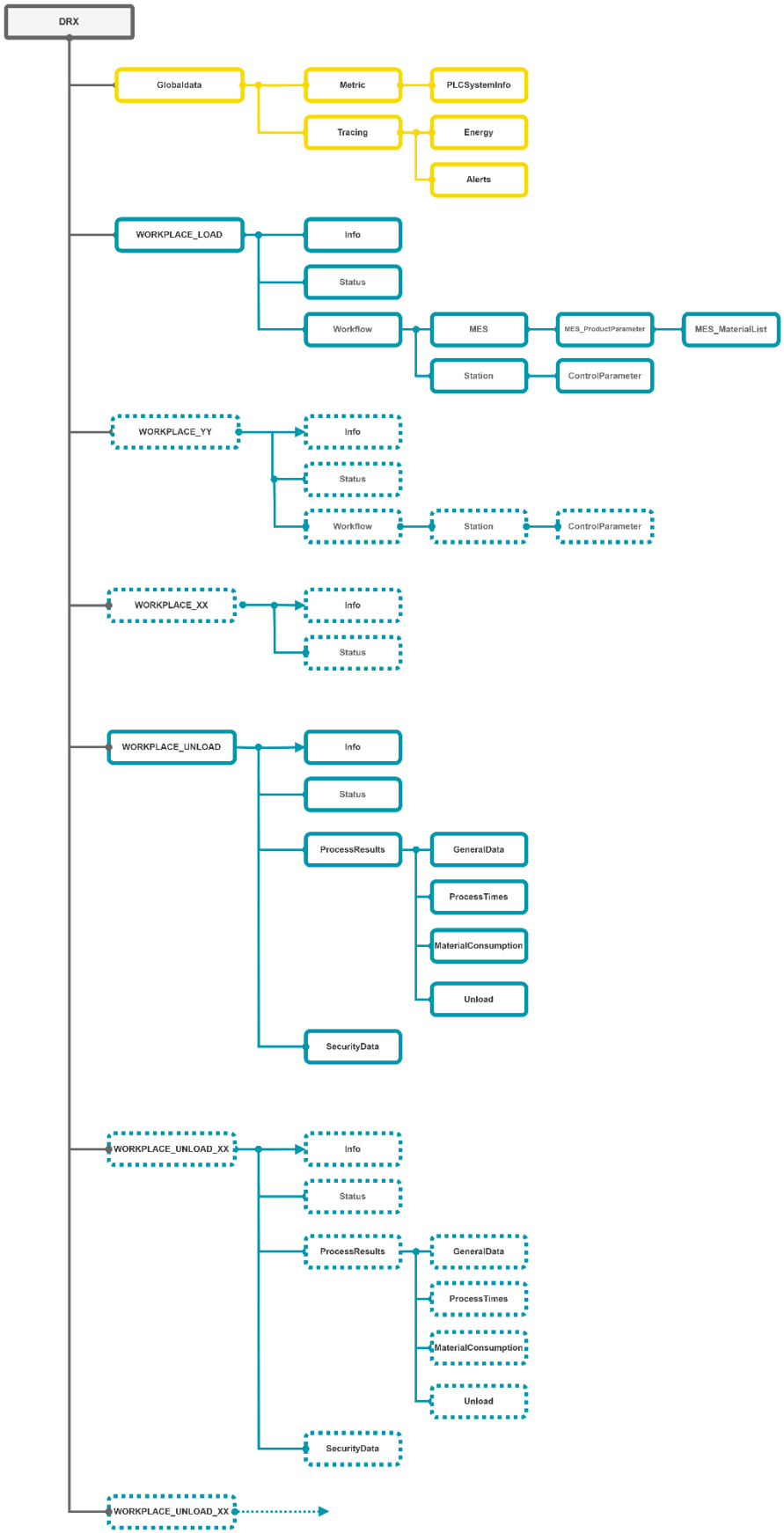


Figure 1: A schematic representation of the information model

The information model uses the root node `DRX` and is basically divided into two main areas:

- Globaldata (The yellow path)
- Workplaces (The turquoise path)

The `Globaldata` section contains nodes which are valid for the entire assembly line (e.g. energy consumptions, alerts).

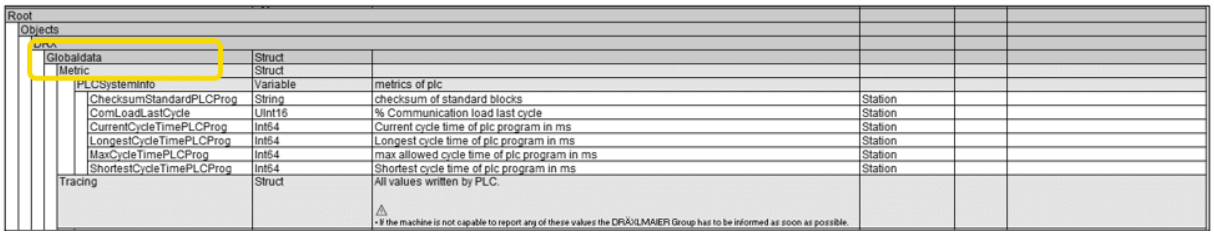


Figure 2: The Globaldata node in the OPC UA information model

The section for workplaces contains data points which are specific for a single process in the assembly line. The total number of used workplaces depends on the complexity of the assembly line, but for sure each machine interface contains at least one Workplace section for the main process. A workplace section is always named `WORKPLACE_XX`:

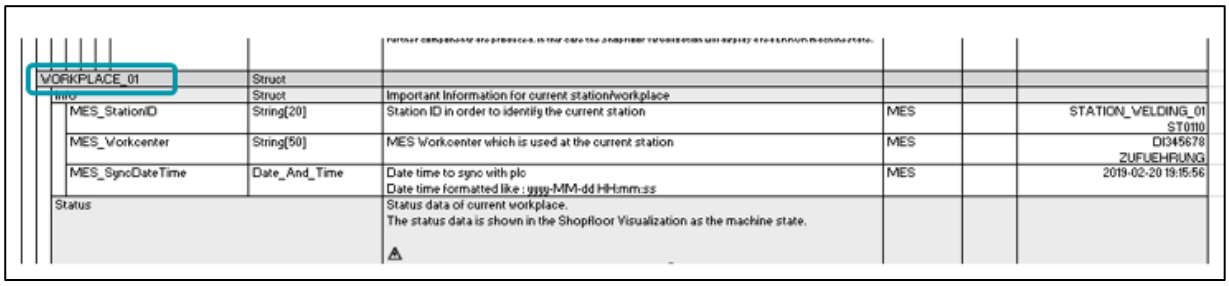


Figure 3: Example of a WORKPLACE node in the OPC UA information model

5.2 Variability

- Some parts of the described information model are variable.
- The final exact information model will be developed during the project.
- Typical variable elements are:

  - The workplace concept (One-Piece-Flow Station vs. Several Automatic Cells)
  - Project specific MES input data (e. g. DMCs or revisions of PCBs)
  - Traceability- and process data points
  - The amount of required WORKPLACE_UNLOAD sections (depending on the amount of data and the requested cycle time)

- During the project phase you will receive a OPC UA nodeset XML file with the exact definition of the information model.
- Get in contact with your DRÄXLMAIER contact person to get more details about this topic.

5.3 Nodeset XML file

The information model is also available as a OPC UA nodeset XML file. You can use this nodeset XML file to import the DRÄXLMAIER OPC UA information model into your development environment (e. g. Siemens TIA).

Note

- Consider the usage of Siemens OPC UA Modeling Editor (SiOME) to integrate the nodeset file into your Siemens TIA Project.
- A short introduction to Siemens SiOME can be provided by the DRÄXLMAIER Group.

The nodeset XML file will be developed during the project phase. It will contain the exact definition of the information model.

5.4 Content

5.4.1 Globaldata

The global data section contains the following two data blocks:

- Globaldata.Metric.PLCSystemInfo
- Globaldata.Tracing

5.4.1.1 PLCSystemInfo

The `Globaldata.Metric.PLCSystemInfo` section is used to monitor the status of a single PLC.

Metric	Struct	The metric section is used to monitor the status of a CPU. For Siemens TIA there are standard functions available to get the values for some of these variables. ⚠ • If the machine is not capable to report any of these values the DRÄXLMAIER Group has to be informed as soon as possible.			
PLCSystemInfo	Variable	metrics of plc			
ChecksumStandardPLCProg	String[18]	checksum of standard blocks	Station		
ComLoadLastCycle	UInt16	% Communication load last cycle	Station		
CurrentCycleTimePLCProg	Int64	Current cycle time of plc program in ms	Station		
LongestCycleTimePLCProg	Int64	Longest cycle time of plc program in ms	Station		
MaxCycleTimePLCProg	Int64	max allowed cycle time of plc program in ms	Station		
ShortestCycleTimePLCProg	Int64	Shortest cycle time of plc program in ms	Station		

Figure 4: Example of the Globaldata.Metric.PLCSystemInfo section

5.4.1.2 Tracing

The `Globaldata.Tracing` section is used to monitor the status of the assembly line.

The following topics will be observed:

- Equipment identification (equipment number and machine number)
- Date and time (synchronized with `MES_SyncDateTime`)
- Lifecycle management of the assembly line software (checksum and version information)
- Global part counters (OK, NOK and SCRAP part counters)
- Global time counters (running time, maintenance time)
- Machine alerts (error code, severity, error description)
- Energy consumption (electricity, air, gas etc.)

AlertActive	Bool	Flag to signalize if an error is active (global level)	Station		true/false
DateTime	DateTime (Siemens: LDT)	Current Date and Time on PLC. Affected by: `MES_SyncDateTime` Date time formatted like : yyyy-MM-dd HH:mm:ss	Station		2019-02-20 19:15:56
EquipmentExemplarNumber	String[18]	A static value, provided by the DRÄXLMAIER Group. The required value can be taken from one of the following DRÄXLMAIER systems: - PLM: Ressource number (starting with 5000) - Host: Optimain number 3 digits (added to Main Number) - SAP: 18 digits - only 10 used (unique when combined with Main Number)	Station		PLM: 5000000823-001 Optimain: 001
EquipmentMainNumber	String[13]	A static value, provided by the DRÄXLMAIER Group. The required value can be taken from one of the following DRÄXLMAIER systems: - PLM: Construction type (starting with 3000) - Host: Optimain number 5 digits - SAP: 13 digits DRÄXLMAIER material number ZFHM	Station		PLM: 300010752-001 Optimain: 87245
InterfaceVersion	String[5]	Version of InterfaceDescription provided by DRX, to be filled by machine manufacturer (according to state of implemetation)	Station		1.00
ResetDate	DateTime (Siemens: LDT)	Date when "total" values have last been reset Date time formatted like : yyyy-MM-dd HH:mm:ss	Station		2019-02-12 12:23:20
SoftwareChecksum	Array[0..x] of String[50]	Hash Checksum of each PLC ⚠ • For normal CPUS using Siemens TIA the checksum can be calculated with GetChecksum function. • For F-CPUS the F-Programsignature should be used	Station		26 5F 6B 9A 5B 01 1D 34
SoftwareVersion	String[20]	The version of the machine software. The format of the version is not fixed. ⚠ • This is not the version of the used Siemens TIA portal! This is the version of the machine software created with Siemens TIA portal.	Station		2017-10-3
TotalMaintenanceTime	Int32	Total time in maintenance (hours) (from reset date)	Station	h	5
TotalNOKParts	Int32	Total amount of NOK parts produced on machine (from reset date); Sum of all stations	Station	pcs	10
TotalOKParts	Int32	Total amount of OK parts produced on machine (from reset date); Sum of all stations	Station	pcs	10
TotalProducedParts	Int32	Total amount of parts produced on machine (from reset date); Sum of all stations	Station	pcs	300
TotalRunningTime	Int32	Total running hours (hours with power supply)	Station	h	2350
TotalScrapParts	Int32	Total amount of scrapped parts produced on machine (from reset date); Sum of all stations	Station	pcs	10

Figure 5: Example of the Globaldata.Tracing section

It also contains the following subsections:

- Globaldata.Tracing.Energy
- Globaldata.Tracing.Alerts

5.4.1.2.1 Energy

The section `Globaldata.Tracing.Energy` is used to report the energy consumption to the DRÄXLMAIER Shopfloor Systems.

If the assembly line is not able to report a required parameter of this section, please get in contact with your contact person of the DRÄXLMIAER Group.

Energy	Struct	Container for all energy monitoring data, dependent on the sensors built in in the machine			
Electricity	Array [0..x] of Struct				
MeterId	String[20]	Identifier of the meter, to enable logical assignment to a physical substation (station ID)	Station		
ActiveEnergy	Float	Total consumed energy since reset	Station	Wh	
ActivePower	Float	Current consumed energy	Station	W	
ApparentPower	Float		Station	VA	
ReactiveEnergy	Float		Station	varh	
ReactivePower	Float		Station	var	
PowerFactor	Float	CosPhi	Station		
CompressedAir	Array [0..x] of Struct				
MeterId	String[20]	Identifier of the meter, to enable logical assignment to a physical substation (station ID)	Station		
Pressure	Float		Station	bar	
FlowRate	Float		Station	Nm³/h	
Count	Float		Station	Nm³	
Gas	Array [0..x] of Struct				
MeterId	String[20]	Identifier of the meter, to enable logical assignment to a physical substation (station ID)	Station		
FlowRate	Float		Station	Nm³/h	
Count	Float		Station	Nm³	
Cooling	Array [0..x] of Struct				
MeterId	String[20]	Identifier of the meter, to enable logical assignment to a physical substation (station ID)	Station		
ActiveEnergy	Float		Station	Wh	
ActivePower	Float		Station	W	
Heat	Array [0..x] of Struct				
MeterId	String[20]	Identifier of the meter, to enable logical assignment to a physical substation (station ID)	Station		
ActiveEnergy	Float		Station	Wh	
ActivePower	Float		Station	W	
Water	Array [0..x] of Struct				
MeterId	String[20]	Identifier of the meter, to enable logical assignment to a physical substation (station ID)	Station		
FlowRate	Float		Station	m³/h	
Count	Float		Station	m³	
Environment	Array [0..x] of Struct				
MeterId	String[20]	Identifier of the meter, to enable logical assignment to a physical substation (station ID)	Station		
LightIntensity	Float		Station	lux	
Temperature	Float		Station	°C	
Humidity	Float		Station	%	
OperatingHours	Float		Station	h	

Figure 6: Example of the Globaldata.Tracing.Energy section

5.4.1.2.2 Alerts

The section `Globaldata.Tracing.Alerts` is used to inform the DRÄXLMAIER Shopfloor Systems that a machine error has occurred on the assembly line (safety door open, no material available etc.). This type of information is called “Alerts”.

These alerts are organized in so called functional groups (`FunctionalGroups`). The task of the functional groups is to group the alerts into logical units.

Example:
All alerts from station 10 will be collected in the functional group “MachineErrorsStation10”.

These functional groups can be freely defined by the machine vendor in consultation with the DRÄXLMAIER Group. Just make sure to use a static, unique and understandable name for each functional group.

For each active alert a code (`AlertCode`), an English message (`AlertMessage`) and a severity of the machine error (`AlertLevel`) must be sent. Inactive alerts must be marked with `AlertCode = -1`, `AlertLevel = -1` and `AlertMessage = “”`.

Alerts	Struct	Structure for transfering detailed information about alerts			
FunctionalGroups	Array [0..x] of Struct	Struct for each functional group in station			
FunctionalGroupName	String[20]	Name of functional group	Station		Workplace_XX_Roboter_01
AlertActive	Bool	Flag to signalize if an alert is active (on functional group level)	Station		true/false
AlertDetails_Count	Int16	Amount of actually filled elements in AlertDetails array	Station		10
AlertDetails	Array [0..x] of Struct	Details for every alert on machine			
AlertCode	Int32	Code of an active alert, to be defined by manufacturer.	Station		[45,23,0,0]
AlertLevel	Int16	Severance of the alert: 1 = ERROR 2 = WARNING 3 = INFO ⚠ • The moment for setting an Alert with AlertLevel = ERROR is not fixed but depends on the machine and must be discussed with the supplier. • Setting an alert with AlertLevel = ERROR will cause the Shopfloor Visualization to display the machine state as ERROR. In this case, all machine states sent from the line within the variables `Idle`, `Running` etc. are not taken into account for displaying. • Setting an alert with AlertLevel = WARNING or AlertLevel = INFO will not cause the Shopfloor Visualization to change the displayed machine state to ERROR. In this case, the displayed machine state in the Shopfloor Visualization depends on the variables `Idle`, `Running` etc.. **Example:** An alert with AlertLevel = ERROR is set at the moment an error on the line becomes active and no further components are produced. In this case the Shopfloor Visualization will display a red ERROR machine state.	Station		[1,2,0,0]
AlertMessage	String[100]	Textual description of the active alert. Matching the code in AlertCode.	Station		[“Printer is not responding (Check power/connection)“, “Roundtable Drive Profinet Connection Error“,,]

Figure 7: Example of the Globaldata.Tracing.Alerts section

5.4.2 Workplace

A `Workplace` section is used to control and monitor a single process. It contains the following sub sections:

- Workplace.Info
  The section `Info` contains static information about the specific workplace (station ID, work center name …).
- Workplace.Status
  The section `Status` contains status information of the process (running, idle, produced parts,…).
- Workplace.Workflow
  The section `Workflow` allows MES to control the process (provide logistic data, provide the BOM …).
- Workplace.ProcessResults
  The section `ProcessResults` is used to report the results of the production back to the DRÄXLMAIER Shopfloor Systems (process- and traceability data, material consumption …).

5.4.2.1 Info

The `Workplace.Info` section contains information about the specific workplace. All data points in this section will be provided by the DRÄXLMAIER MES.

Info	Struct	Important Information for current station/workplace			
MES_StationID	String[20]	Station ID in order to identify the current station	MES		STATION_WELDING_01 ST0110
MES_SyncDateTime	DateTime (Siemens: LDT)	Date time to sync with plc Date time formatted like : yyyy-MM-dd HH:mm:ss	MES		2019-02-20 19:15:56
MES_Workcenter	String[50]	MES Workcenter which is used at the current station	MES		DI345678 ZUFUEHRUNG

Figure 8: Example of the Workplace.Info section

5.4.2.2 Status

The `Workplace.Status` section contains live status information of the workplace (machine running, machine idle, current order/product in place in the station, parts produced on this workplace …).

Status	Struct	Status data of current workplace. The status data is shown in the Shopfloor Visualization as the machine state. ⚠ • If the machine is not capable to report any of these values the DRÄXLMAIER Group has to be informed as soon as possible.			
Active	Bool	False: station/turntable/control side deactivated or not available True: station/turntable/control side activated	Station		true/false
AutomaticMode	Bool	False: Plc in manual mode, not controlled by mes system True: Plc controlled by mes system	Station		true/false
Equipping	Int16	0: Machine is currently not equipping 1: Machine is beeing equipped with new tools -1: If the machine is not capable to report the equipping mode.	Station		0
Idle	Bool	False: PLC not ready for MES_Release. PLC is not ready to start production (is running, in maintenance,...) True: PLC ready for next part to produce ⚠ • If the production line supports only one machine state (one state for the whole production line) make sure to use the same state in WORKPLACE_UNLOAD.Status.Idle.	Station		true/false
LastCycleTime	Float	Cycle time for the current part from start to end of production (machine time + human time) Measurement start: 1st activity which can be detected by machine Measurement end: last activity which can be detected by machine	Station		23.40
MaintenanceMode	Int16	0: Machine is currently running on normal mode 1: Machine is running in maintenance mode -1: If the machine is not capable to report the maintenance mode.	Station		1
Running	Bool	Affected by: `MES_Release` False: PLC production not in progress True: Production in progress, part is processed	Station		true/false
Sampling	Int16	0: machine is currently not in sampling mode 1: machine is running test/teaching parts -1: machine is not capable to report the sampling mode since it is not available.	Station		-1
TotalNOKParts	Int32	Number of NOK parts on this specific workplace. ⚠ • NOK parts can be reworked.	Station		0
TotalOKParts	Int32	Number of OK parts on this specific workplace	Station		1
TotalScrapParts	Int32	Number of SCRAP on this specific workplace ⚠ • SCRAP parts can not be reworked.	Station		0
ProductInformation	Array [0..x] of Struct	Information about all active products in current workplace. ⚠ • This information is mandatory also if the cycle time is quite low.			
ProductID	String[100]	Currently processed ProductID (based on `MES_ProductID`).	Station		BM2F094126CA3320; 300000220-001P001007827
ProductType	String[50]	Currently processed Product Type (based on `MES_ProductType`).	Station		BM2F094112;300000220-001
Status	Int16	0: not available 1: WAITING 2: RUNNING 3: CANCELLED 4: INTERRUPTED	Station		1

Figure 9: Example of the Workplace.Status section

5.4.2.3 Workflow

In this section all variables for fulfilling a specific workflow for a certain workplace are provided. It contains the following subsections:

- Workplace.Workflow.MES
- Workplace.Workflow.Station

5.4.2.3.1 MES

In this `Workplace.Workflow.MES` section DRÄXLMAIER MES will provide all needed logistic information to produce a new order of products. This set of data is often called “order data”:

MES_Action	String[20]	START, CANCEL, REORDER	MES		SCAN, WELD, CANCEL,CRIMP,..
MES_CancelProcess	Bool	False: Process not canceled True: Process canceled due to any error ⚠ • This value is typically not used by production lines in the segment CS (if they are using only to 2 workplaces).	MES		true/false
MES_DataProcessed	Bool	Signals if result has been read and processed Affected by: `DataAvailable` False: MES system has not processed data True: MES system has read processdata. `DataAvailable` must be set to 0	MES		true/false
MES_PartsToProduce	Int32	Total amount of parts to be produced for the current job ⚠ • Can count up to 999999	MES		e.g. 234897 or e.g. 21
MES_ProductID	String[100]	The ProductID of the requested job. Used to identify the requested product for this new job. Used to link the process data to this new job. ⚠ • See the chapter "ProductID" for a detailled explanation of the used format.	MES		300000220-001P001007827000001; BM2F094126CA3320; E9904969P40102
MES_ProductType	String[50]	The ProductType of the requested job.	MES		300000220-001; BM2F094112; P401
MES_Release	Bool	production release False: production not allowed, don't start machine True: start production	MES		true/false

Figure 10: Example of the Workplace.Workflow.MES section

The section `MES` also contains the subsection `MES_ProductParameter`: In this section product and process related parameters are transferred from the MES to the assembly line. This set of parameters is also called “the recipe” and contains e.g. the BOM within the `MES_MaterialList` structure:

MES_ProductParameter	Struct	These variables are usually added when additional input parameters from MES to the machine are required.			
MES_CustomerChangeIndex	String[6]	Customer change index ⚠ • The format of the customer change index (from MES field "ordInfoField3") is not fixed. • Get in contact with the DRÄXLMAIER group to get the latest example.	MES		01S
MES_CustomerChangeLevel	String[15]	Customer change level ⚠ • The format of the customer change level (from MES field "ordInfoField2") is not fixed. • Get in contact with the DRÄXLMAIER group to get the latest example.	MES		22.10.2015
MES_CustomerNumber	String[20]	Customer number ⚠ • The format of the customer number (from MES field "ordInfoField1") is not fixed. • Get in contact with the DRÄXLMAIER group to get the latest example.	MES		7641798-03
MES_JobNr	Int16	Can be used to define different product variants that have to be handled on machine side 0: not available (To be defined)	MES		1
MES_PackagingType	String[15]	Box ID (Behälter ID from MES)	MES		APE1
MES_StartSinglePartID	Int32	Singlepartnumber for starting the numeration on the machine. At the first start usually "1", after restarting an order this number will contain the last number from the result data incremented by 1. ⚠ • Can count up to 999999	MES		992330
MES_MaterialList	Array[0..x] of Struct	List of needed materials. The format of the list is project specific. Please get in contact with DRX to	MES		
MES_MaterialNr	String[100]	Material number of current material	MES		300000220-001; T0101589
MES_BoxNr	Int16	Box number of current material	MES		2
MES_Handworkplace	Int16	Number of the desired hand-workplace fur current material	MES		5

Figure 11: Example of the Workplace.Workflow.MES.MES_ProductParameter section

Note

- Previously, the contents of the MES_MaterialList section were also referred to as the "IPST parameter list (IPST-Parameterliste)".

`MES_ProductParameter` also contains a space in which project specific parameters can be transferred from the MES to the assembly line. If required, these parameters will be defined during the project phase.

MES_ToBeDefined		Placeholder for other additional variables. ⚠ • These variables are optional and not mandatory. • These additional variables must be agreed between DRÄXLMAIER and the manufacturer.	MES		e.g. FazitString

Figure 12: Project specific parameters within Workplace.Workflow.MES.MES_ProductParameter

5.4.2.3.2 Station

In the `Workplace.Workflow.Station` section the workplace reports the status of the workflow (e.g. result, data available …).

DataAvailable	Bool	False: data not available or not related to current production process True: data available and actual. Must not be set to 0 until `MES_DataProcessed` is set to 1	Station		true/false
Result	Int16	Result of loading process 0: not available 1: OK / INPRODUCTION 2: NOK 3: CANCEL	Station		2

Figure 13: Example of the Workplace.Workflow.Station section

It also contains the subsection ControlParameter:

The optional `ControlParameter` section is used to transfer nonstandard information from the assembly line to the MES. A typical example is the result of a scanned barcode. If required, this nonstandard information is defined during the project phase.

ControlParameter	Struct	Placeholder for needed values which are sent by the machine to the MES			
ToBeDefined		Placeholder for other additional variables. ⚠ • These variables are optional and not mandatory. • These additional variables must be agreed between DRÄXLMAIER and the manufacturer.	Station		

Figure 14: Example of the optional Workplace.Workflow.MES.ControlParameter section

5.4.2.4 SecurityData

The section `Workplace.SecurityData` is used to report results of cyber security related manufacturing processes to the DRÄXLMAIER Shopfloor Systems. Currently two different cyber security related manufacturing processes are supported:

- Locking a debug interface by password
- Certificate signing request

SecurityData	Struct	Placeholder for security relevant data which should not be visisble in the data structure of processresults. Structure should only be existing if it is in use for this machine/project ⚠ • These variables are optional and not mandatory. • These additional variables must be agreed between DRÄXLMAIER and the manufacturer.			
ProductID	String[100]	Currently processed ProductID (based on MES_ProductID).	Station		Format MES/HOST: BM2F094112CA3320 Format MES2/SAP: 300000220-001P001007827
CSR	String	CSR String of the product ⚠ The String length of this variable could expand a lot. Maximum String length will be defined within the project.	Station		
PublicKey	String	Public key of the product (base64 encoded DER) ⚠ The String length of this variable could expand a lot. Maximum String length will be defined within the project.	Station		
Signature	String	Signature of the product (base64 encoded) ⚠ The String length of this variable could expand a lot. Maximum String length will be defined within the project.	Station		
ToBeDefined		Placeholder for other additional variables. ⚠ • These variables are optional and not mandatory. • These additional variables must be agreed between DRÄXLMAIER and the manufacturer.	Station		

Figure 15: Example of the optional Workplace.SecurityData section

5.4.2.5 ProcessResults

The section `Workplace.ProcessResults` is used to report the results of the production back to the DRÄXLMAIER Shopfloor Systems.

All the project specific process- and traceability data points will be added underneath the `ProcessResults` section. The data points required are process dependent and are therefore defined as part of the project. Due to this circumstance, this section of the interface is marked with ToBeDefined in this template.

ToBeDefined		Placeholder for the process parameters of the finished part belonging to the predefined processes for this station. The complete list of process parameters is provided by the DRÄXLMAIER group.	Station		

Figure 16: Example process specific data in the information model

Note

- If you need the required project specific process- and traceability data earlier, please contact your responsible contact person. He or she can provide you with the process specific nodesets.

The section contains the following sub sections:

- Workplace.ProcessResults.GeneralData
- Workplace.ProcessResults.ProcessTimes
- Workplace.ProcessResults.MaterialConsumption
- Workplace.ProcessResults.Unload

5.4.2.5.1 GeneralData

The section `Workplace.ProcessResults.GeneralData` is used to report the amount of produced OK, NOK or SCRAP parts together with detailed Error information for those special parts in case any error occurred.

GeneralData					
ErrorcodesNOK	Array [0..x] of String[30]	Error codes for each part that was sent as NOK from machine. ⚠ • The error codes have to be defined by the machine manufacturer. • The error codes have to use the format specified in chapter "Format of an Error Number".	Station		[2,0,0,0]
ErrorcodesScrap	Array [0..x] of String[30]	Error codes for each part that was sent as Scrap from machine ⚠ • The error codes have to be defined by the machine manufacturer. • The error codes have to use the format specified in chapter "Format of an Error Number".	Station		[4,0,0,0]
ErrormessagesNOK	Array [0..x] of String[150]	Textual description of an error code from field ErrorcodesNOK ⚠ • One message for each error code in `ErrorcodesNOK` • Errormessage(s) should be collected for each part and have to be deleted after each cycle (MES_DataProcessed=1) • The content may not contain special characters that are also used within an XML notation, e.g. &, <, >	Station		[Torque out of limit,,,]
ErrormessagesScrap	Array [0..x] of String[150]	Textual description of an error code from field ErrorcodesScrap ⚠ • One message for each error code in `ErrorcodesScrap` • Errormessage(s) should be collected for each part and have to be deleted after each cycle (MES_DataProcessed=1) • The content may not contain special characters that are also used within an XML notation, e.g. &, <, >	Station		[Surface too hot,,,]
PartsNOK	Int32	Number of NOK parts for current process. ⚠ • NOK parts can be reworked.	Station		0
PartsOK	Int32	Number of OK parts for current process.	Station		1
PartsScrap	Int32	Number of SCRAP parts for current process.. ⚠ • SCRAP parts can not be reworked.	Station		0
ProductID	String[100]	Currently processed ProductID (based on `MES_ProductID`). ⚠ • See the chapter "ProductID" for a detailled explanation of the used format.	Station		BM2F094126CA3320;300000220-001P001007827
ProductType	String[50]	Currently processed Product Type (based on `MES_ProductType`).	Station		BM2F094112;300000220-001
AlternativeID	Array [0..x] of String[100]	Barcode (e. g. DMC, Code128, ...) on the final part, that is/are different from the MES internal ProductID and that might be needed for proper tracking of process data in MES. Used to link the MES internal part data to the physical machine part data. ⚠ • The content of the AlternativeID is not fixed and is optional to be filled. • Get in contact with the DRÄXLMAIER group for a final definition if the filling is needed and which value should be filled in.	Station		BUC13092023111299233000022O-001P001007827
TotalProcessingTime	Float	Processing time (duration) for all involved actions in total	Station	minutes	1.87

Figure 17: Example of the Workplace.ProcessResults.GeneralData section

5.4.2.5.2 ProcessTimes

The section `Workplace.ProcessResults.ProcessTimes` is used to report detailed start and end times of individual process steps:

Examples:

- Robot picks up a component and puts it down.
- Entry and exit of a product carrier.

These groups can be freely defined by the machine vendor in consultation with the DRÄXLMAIER Group.

ProcessTimes	Array of Struct	process times for each process			
EquipmentID	String[30]	ID of the source which reported this time span (e.g. StationID, ToolID…)	Station		ST010_IR01
OperationID	String[20]	Operation which was performed during that time (e.g. pick, driving in, waiting…)	Station		1.0020-10.10
Description	String[50]	short desciption	Station		
StartTime	DateTime (Siemens: LDT)	Start timestamp	Station		18.05.2020 13:00
EndTime	DateTime (Siemens: LDT)	End timestamp	Station		18.05.2020 13:15

Figure 18: Example of the Workplace.ProcessResults.ProcessTimes section

5.4.2.5.3 MaterialConsumption

The section `Workplace.ProcessResults.MaterialConsumption` is used to report the consumed materials back to the MES and ERP. It is divided into two subsections: `AddtionalConsumption` and `RegularConsumption`.

- `RegularConsumption`: This section must be filled for every part. If the part is an OK part, it must contain exactly the materials of the BOM. If the part is NOK or SCRAP, the section may contain fewer materials than in the BOM.

- `AdditionalConsumption`: This section is only filled if e.g. due to rework more material than planned was used. Only the materials that were additionally used for the rework have to be inserted in this section.

MaterialConsumption	Struct	container for material consumption			
AdditionalConsumption_Count	Int16	Amount of actually filled elements in AdditionalConsumption array	Station		5
AdditionalConsumption	Array of Struct	list of all material commissioned additionally (exeeding the regular consumption)			
MaterialNr	String[100]	Host: Material label SAP VDA: Segment P SAP RML: Pos. 1-13	Station		
Consumption	Float	Consumed amount	Station		
Charge	String[30]	Host: Batch label SAP VDA: Supplier or DRÄXLMAIER internal Batch VDA Segment 1T SAP RML: Pos. 14-23	Station		
ExpirationDate	String[20]	optional SAP VDA: Segment 14D SAP RML: n/a	Station		
DeliveryNr	String[30]	optional SAP VDA: Segment 2S SAP RML: n/a	Station		
HandlingUnit	String[20]	SAP VDA: Segment 1J RML: Pos. 24-43	Station		
LabelContent	String[300]	whole 2D barcode label content for VDA including special characters has to be masked, e.g. **#&30;**	Station		
PickPosition	String[20]	optional SAP storage unit where the part was picked (hardcoded on PLC, eg. 0000-000-000-000)	Station		
PositionIndex	String[20]	Detailed position where the material was mounted in the product (optional, only needed if singlepart tracking is required)	Station		
SinglepartLabelSupplier	String[100]	Unique supplier label on the product for traceability. (optional, only needed if singlepart tracking is required)	Station		
RegularConsumption_Count	Int16	Amount of actually filled elements in RegularConsumption array	Station		5
RegularConsumption	Array of Struct	list of all commissioned material			
MaterialNr	String[100]	Host: Material label SAP VDA: Segment P SAP RML: Pos. 1-13	Station		
Consumption	Float	Consumed amount	Station		
Charge	String[30]	Host: Batch label SAP VDA: Supplier or DRÄXLMAIER internal Batch VDA Segment 1T SAP RML: Pos. 14-23	Station		
ExpirationDate	String[20]	optional SAP VDA: Segment 14D SAP RML: n/a	Station		
DeliveryNr	String[30]	optional SAP VDA: Segment 2S SAP RML: n/a	Station		
HandlingUnit	String[20]	SAP VDA: Segment 1J RML: Pos. 24-43	Station		
LabelContent	String[300]	whole 2D barcode label content for VDA including special characters has to be masked, e.g. **#&30;**	Station		
PickPosition	String[20]	optional SAP storage unit where the part was picked (hardcoded on PLC, eg. 0000-000-000-000)	Station		
PositionIndex	String[20]	Detailed position where the material was mounted in the product (optional, only needed if singlepart tracking is required)	Station		
SinglepartLabelSupplier	String[100]	Unique supplier label on the product for traceability. (optional, only needed if singlepart tracking is required)	Station		

Figure 19: Example of the Workplace.ProcessResults.MaterialConsumption section

5.4.2.5.4 UNLOAD

The `Workplace.ProcessResults.UNLOAD` section contains various general parameters that must be reported for each produced part.

UNLOAD			to be used for specific process parameters that are not assigned to a special process			
	FileName	String[50]	If additional data is transferred through Data2Cloud (via a file interface), this variable should contain the respective file name including the extension. To ensure uniqueness the file name should contain the ProductID plus a time stamp. ⚠ • If file transfer is expected, but the machine is unable to save the file in the designated folder, the machine has to return an ErrorMessage to signalize the problem to MES • If no file is to be saved, the value of this field should be empty	Station		A12345670101_20191205123001.zip
	SinglePartID	String[6]	The part number of the currently processed single part. **Format:** Fixed length of 6 digits, use leading "0" characters if needed.	Station		000001
	QualityPart	Bool	False: Part was produced normally True: Part was produced for QS reasons	Station		true/false
	ToBeDefined		Placeholder for the process parameters of the finished part that don´t belong to a special process	Station		

Figure 20: Example of the Workplace.ProcessResults.UNLOAD section

6 Workplace Concept & Workflows

6.1 Introduction

The information model divides the assembly line into so-called “Workplaces”. The number of used workplaces depends on the complexity of the assembly line. The following pictures show examples of the two most frequently used concepts:

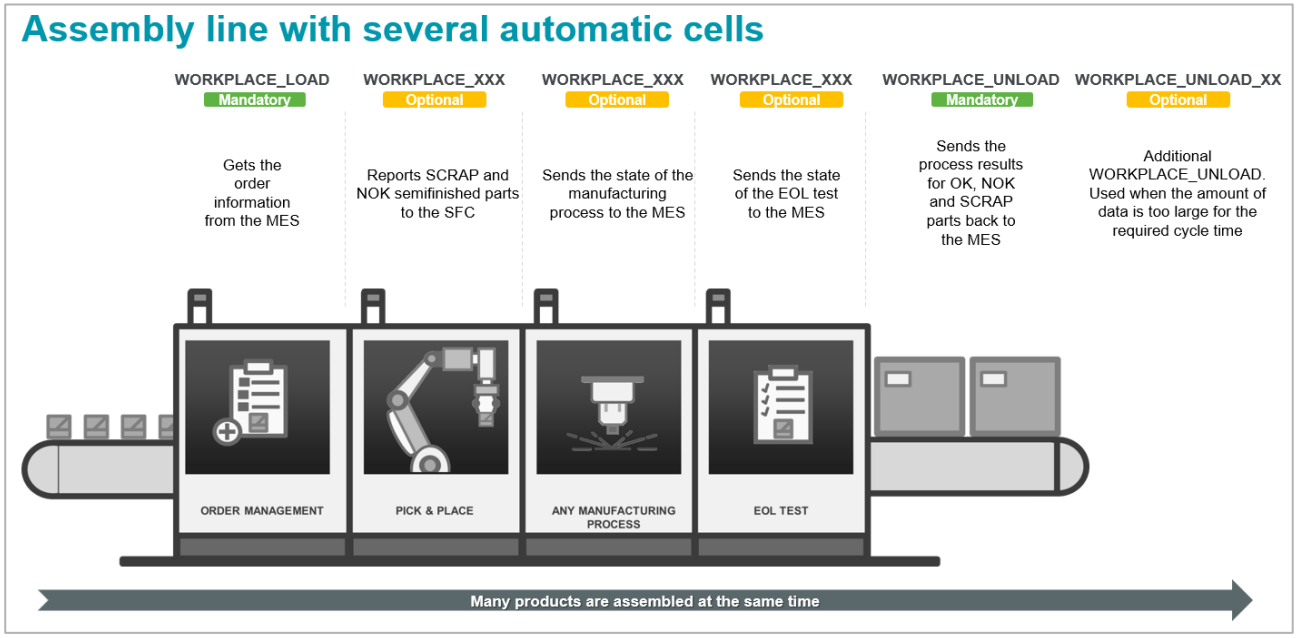


Figure 21: Workplace concept of an assembly line with several automatic cells

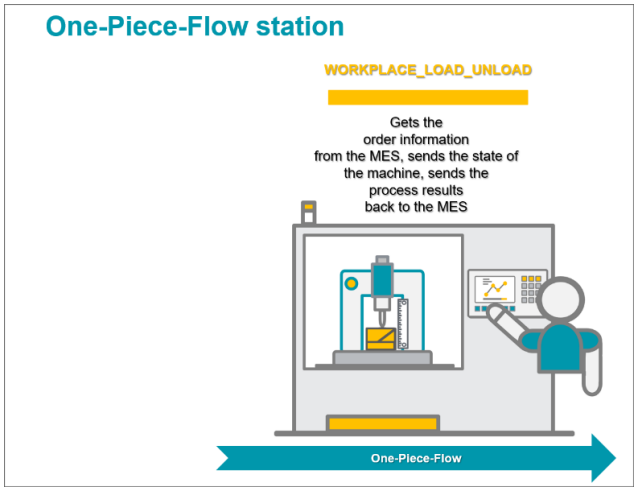


Figure 22: Workplace concept of a One-Piece-Flow assembly line

Each workplace can get a dedicated section in the information model if needed. The transaction-safe communication between each workplace and the MES is done by using predefined workflows.

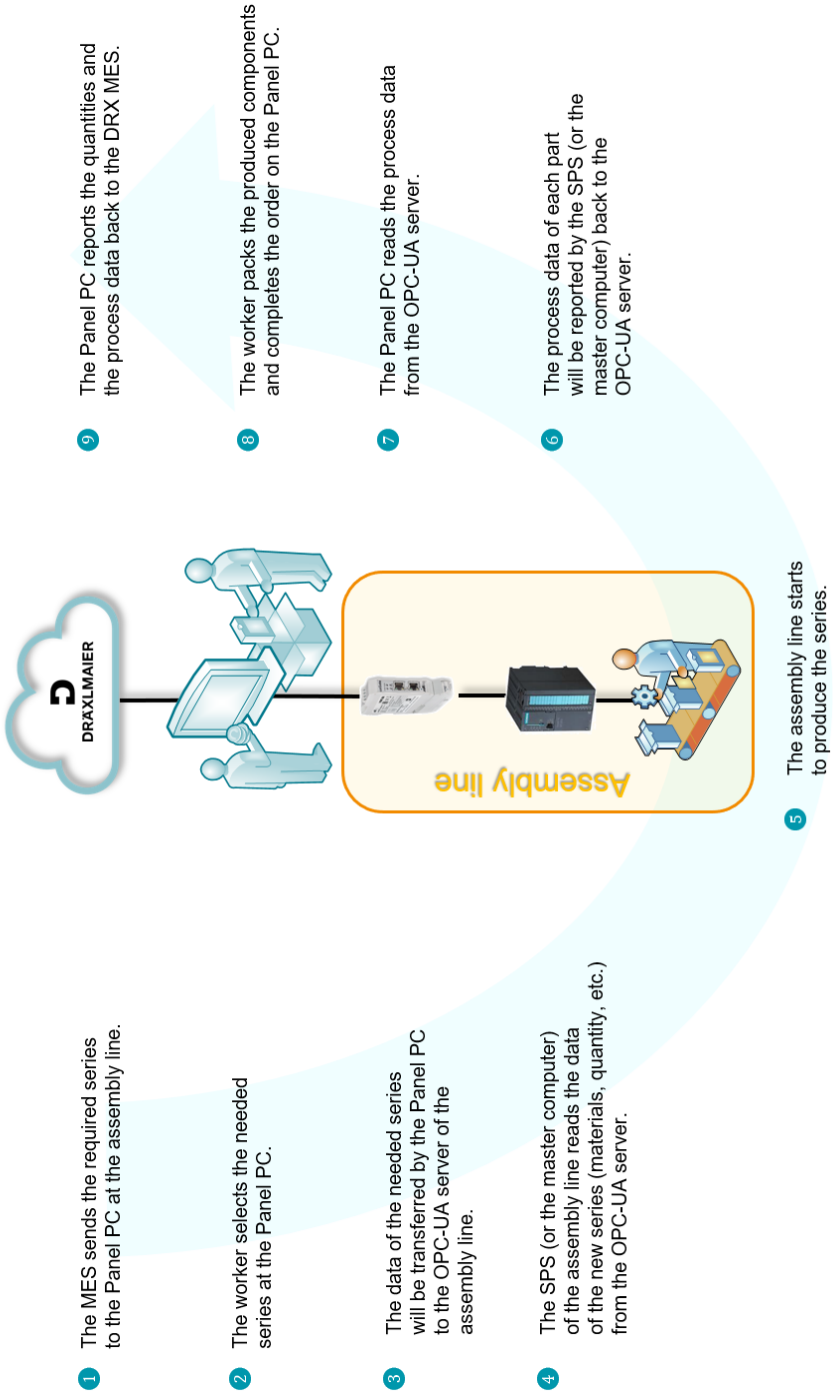
Note

- The configuration of the machine interface depends very much on the type of the assembly line.
- The final workplace configuration is developed within the project.
- Get in contact with the DRÄXLMAIER Group to get detailed information about the used workplace configuration.

6.2 Concept 1: Several Automatic Cells

This concept must be used whenever more than one part is inside the machine at the same time.

6.2.1 Typical Operating Procedure



6.2.2 WORKPLACE_LOAD

The following workflows are needed to start an order on the assembly line in a proper way and therefore will be handled in the `WORKPLACE_LOAD` section in the information model.

Note

- Without further notice all workflows are based on the standard handshake which can be found in the applicable document “InterfaceSpecification&Structure_OPC_UA_template_V3_xx.pdf” in chapter “Standard Handshake”.

6.2.2.1 Workflow “Start Order(s)“

To start an order on the assembly line the MES is starting a workflow with the `MES_Action = "START"`.

The number of orders that can be sent to the assembly line is limited for each machine. If the worker tries to start more orders than the machine can handle there will be an error message on the CAAFrontend.

The assembly line is responsible to hold an internal counter that counts the part according to the sent start number for counting `MES_StartSinglePartID`.

If an order is restarted after a prior cancellation on the CAAFrontend, the number of parts to be produced `MES_PartsToProduce` will be reduced with the parts that were already finished. So only the new calculated amount will be sent to the assembly line.

The assembly line reads out the values and starts the production with the given order data.

Note

- It must be possible to start the assembly line without an order from MES. In case the connection to the MES is broken, please see chapter ”Malfunction of the MES Communication Interface”.

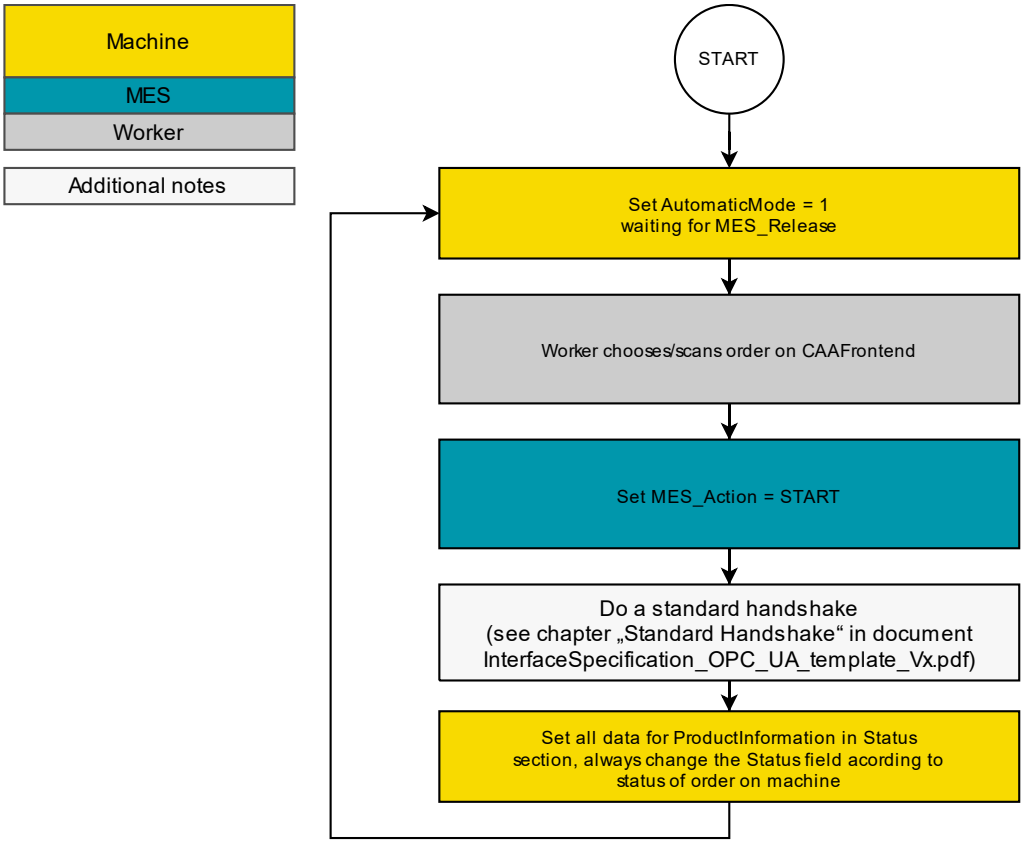


Figure 23: Workflow “Start Order(s)”

6.2.2.2 Workflow “Cancel Order(s)“

To cancel an order on the assembly line the MES is starting a process with MES_Action = “CANCEL”. MES will also communicate the affected order in MES_ProductID.

In this case all parts that are currently started on the assembly line can be finished. No new parts should be allowed to be put on the assembly line by the worker.

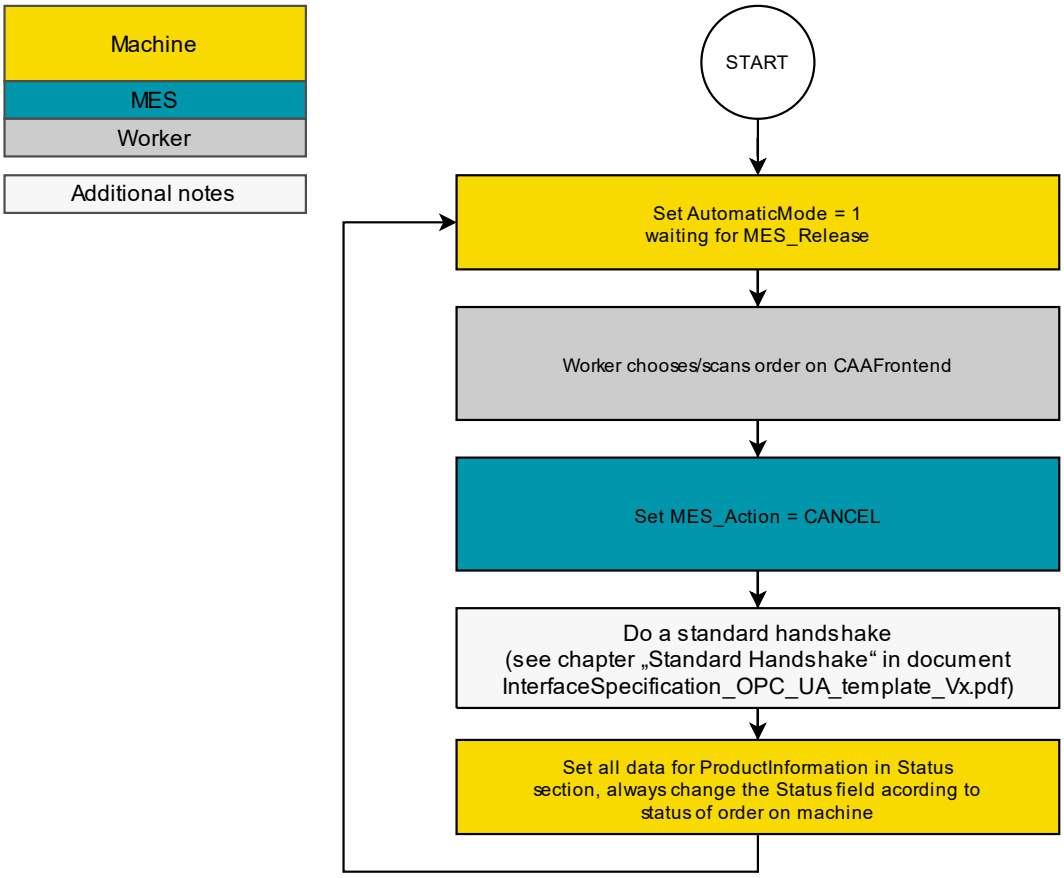


Figure 24: Workflow “Cancel Order(s)”

6.2.2.3 Workflow “Manually Reorder Single Parts“

All parts that are booked as scrap by the worker must be manually reordered at the assembly line. This process is only suitable for parts that were already declared as OK from the assembly line - but scrapped due to any other reason afterwards, e.g., they fell down and broke.

In this case the MES will set the `MES_Action` = “`REORDER`” and reports the affected single part in `MES_ProductID`. The reorder is done for every part separately and for each reordering the value in `MES_PartsToProduce` will be set to 1.

A reorder is only possible if the series/order is still active at the assembly line. If the series is not active, there will be an error message on the CAAFrontend.

If the last part for an order was already inserted in the assembly line, the incrementation must be ignored from the assembly line.

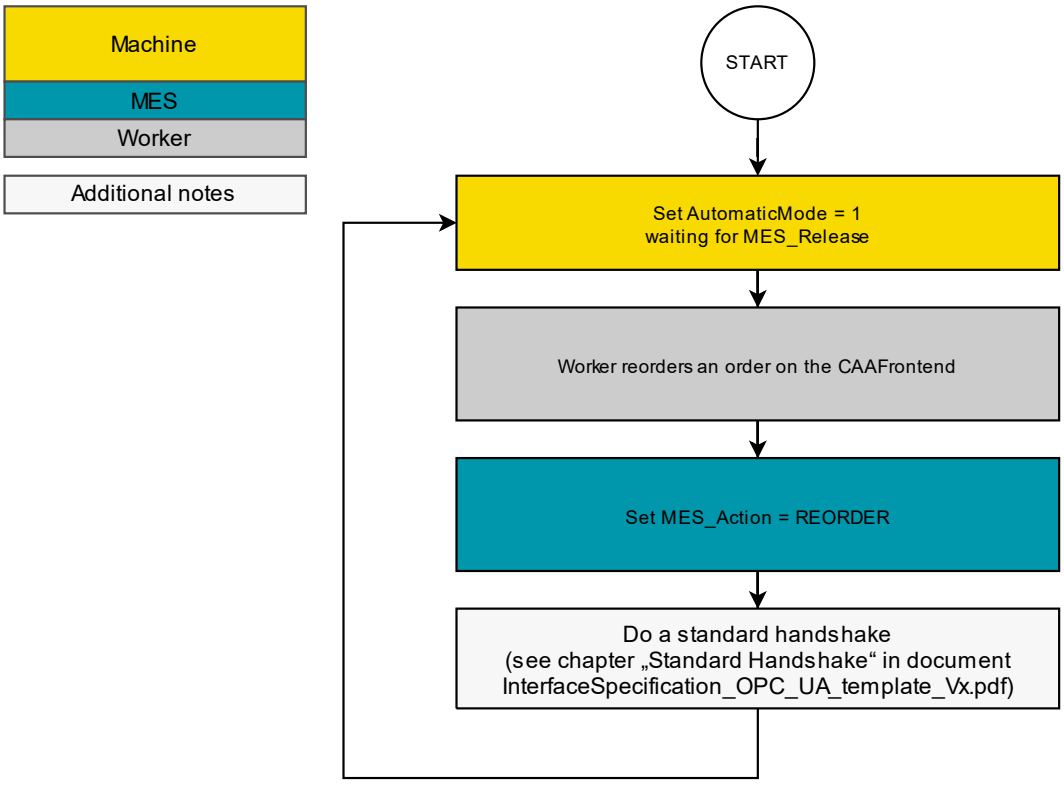


Figure 25: Workflow “Manually Reorder Single Parts“

6.2.3 WORKPLACE_XX

In addition to the two mandatory workplaces WORKPLACE_LOAD and WORKPLACE_UNLOAD, further workplaces can be added as needed.

Those WORKPLACE_XX are providing a detailed representation of the machine state for a single process or automatic cell (e.g. running, idle, maintenance).

The final number of workplaces is defined during the project.

6.2.4 WORKPLACE_YY

In addition to the two mandatory workplaces WORKPLACE_LOAD and WORKPLACE_UNLOAD, further workplaces can be added as needed.

Those WOKRPLACE_YY are providing a special communication between MES and a single process (e. g. password request, special checks for scanned parts in between the WORKPLACE_LOAD und WORKPLACE_UNLOAD).

In the following chapter there is one example for a special checking within the WORKPLACE_CHECK.

6.2.4.1 WORKPLACE_CHECK

In some situations the machine needs to interact with MES to verify, if a special consumed part (e.g. a platine) is allowed to be used at machine inbetween the LOAD and the UNLOAD process.
With the following process the barcode of the used part can be sent to MES in order to check the validity of the part and provide additional data for that part (if needed). The process will be handled on `WORKPLACE_CHECK` datablock in OPC UA server.

In order to start the checking process the field `ProductID` from `GeneralData` section has to be filled and afterwards the `ProductIDAvailable` has to be set to 1.

MES checks the validity of the sent ProductID. If the part is valid, the value of `MES_Action` will be set to "`CONTINUE`". It will be set to "`REPLACE`" in case the part is not valid and has to be replaced by machine. MES will fill any additional data (in case it is required) structure in `MES_ProductParameter`.

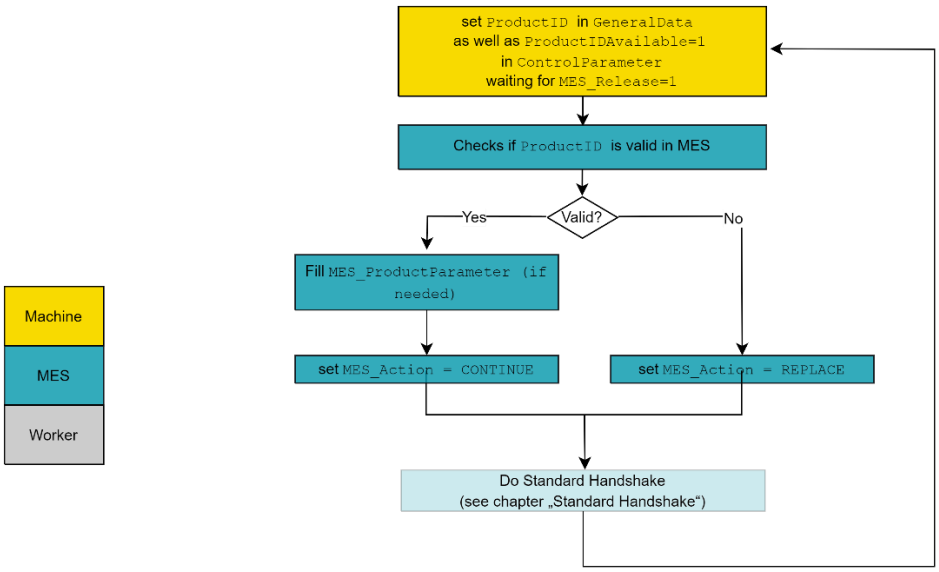


Figure 26: Workflow for WORKPLACE_CHECK

Note

- Contact the DRÄXLMAIER Group to clarify whether the `WORKPLACE_CHECK` is required for your assembly line and if any additional data sent within the `MES_ProductParameter`, has to be evaluated.

6.2.5 WORKPLACE_UNLOAD (WORKPLACE_UNLOAD_XX)

For unloading the parts from the assembly line (sending the process results to the Shopfloor Systems), a `WORKPLACE_UNLOAD` section in the information model must be used.

The number of required `WORKPLACE_UNLOAD` sections depends on the ratio of cycle time to data volume. If the data cannot be transmitted in the specified cycle time, one or more additional `WORKPLACE_UNLOAD_XX` sections are added.

Note

- Without further notice all workflows are based on the standard handshake which can be found in the applicable document “InterfaceSpecification&Structure_OPC_UA_template_V3_xx.pdf” in chapter “Standard Handshake”.
- The actual number of WORKPLACE_UNLOAD sections is finally defined in the project. As a rough guide, additional WORKPLACE_UNLOAD_XX sections could be required for a cycle time of less than 10 seconds.

6.2.5.1 Workflow “Provide Result Data“

As soon as a part is completed on the assembly line, the result for OK, NOK and SCRAP parts (including process data) must be transmitted to the MES.

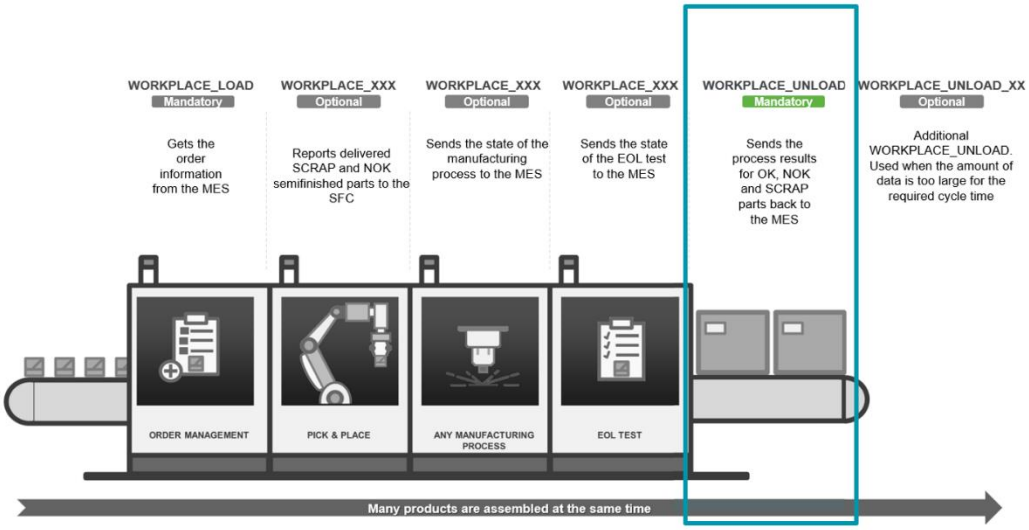


Figure 27: Reporting of OK, NOK and SCRAP part results to MES

The DRÄXLMAIER Shopfloor Systems distinguish between three different part statuses OK, SCRAP and NOK:

- OK means that the component has been produced without defects.
- NOK means that the production of the component is not yet complete and further production steps are necessary (e. g. rework, run-in etc.).
- SCRAP means that the component was manufactured incorrectly and must be scrapped. Reworking is no longer possible.

Important

- It must be assured that the data is completely written to the OPC UA server (all parameters are up to date) before the CAAFrontend is informed by setting `DataAvailable` to 1.
- It must be assured that all nodes belonging to the OPC structure `ProcessResults` have been cleared and initialized by the PLC before new process data is written to it. See the chapter “8 Initializing process- and traceability data”.
- Data of finished parts can be provided in chaotic way. So, it is possible that part 000003 will be booked before part 000002. However, the numbering must be consecutive at the end of the series without any gaps! Only in the case that the worker had to remove a complete product carrier (e.g. caused by a crash) gaps are allowed.
- If a worker takes out a part from the assembly line, there must be a possibility to write the part manually NOK or SCRAP inside the assembly line.

- In order to reach the number of needed parts, all parts that are NOK or SCRAP on the assembly line during production have to be reordered automatically by the assembly line (by increasing the amount by 1 for each NOK or SCRAP part).

  This is only valid as long as there are open quantities available for this order. If the order was already completed at the assembly line and some of the parts that are currently in the line will be NOK or SCRAP, then no reorder should be done for these parts by the assembly line. Due to this it might be possible that the needed number of OK parts will not be reached in the end and there will be an underdelivery to the customer.

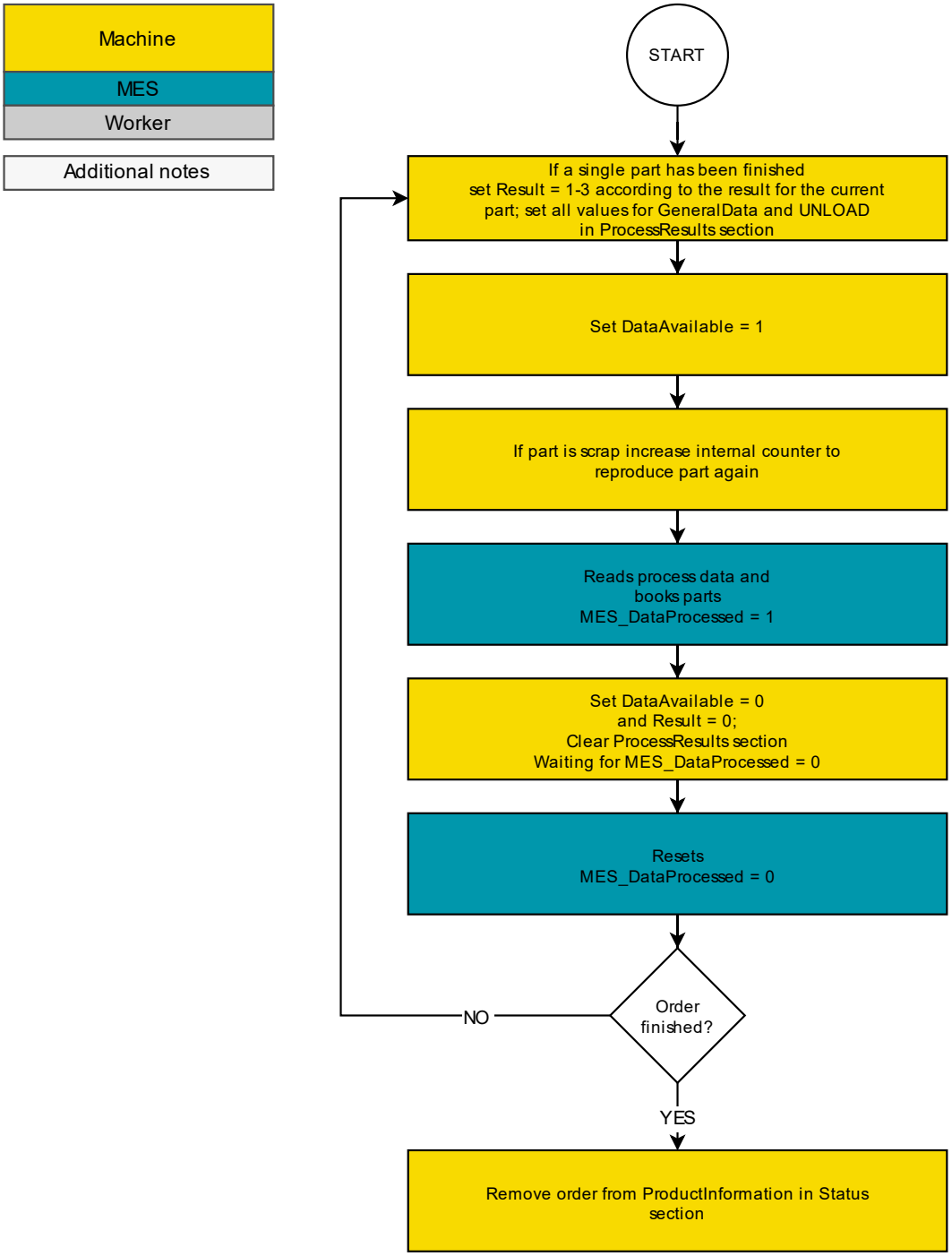


Figure 28: Workflow "Provide Result Data"

6.2.5.2 Workflow “Interrupt Process”

If an unexpected communication error occurs (between MES and assembly line) the entire process must be interrupted.

Typical reasons for such communication errors are:

- Duplicated part numbers: If a part number has already been used in the past.
- Booking problems in MES: database problems, missing settings, maximum quantity reached etc.

If such a problem occurs, MES will set `MES_BookingError=1` (instead of `MES_DataProcessed=1`). Also the variable `MES_BookingErrorText` will be filled with a description of the error by MES.

The content of `MES_BookingErrorText` and a button to skip the affected part should be displayed on the machine HMI.

The following figure shows an example of such a machine HMI page:

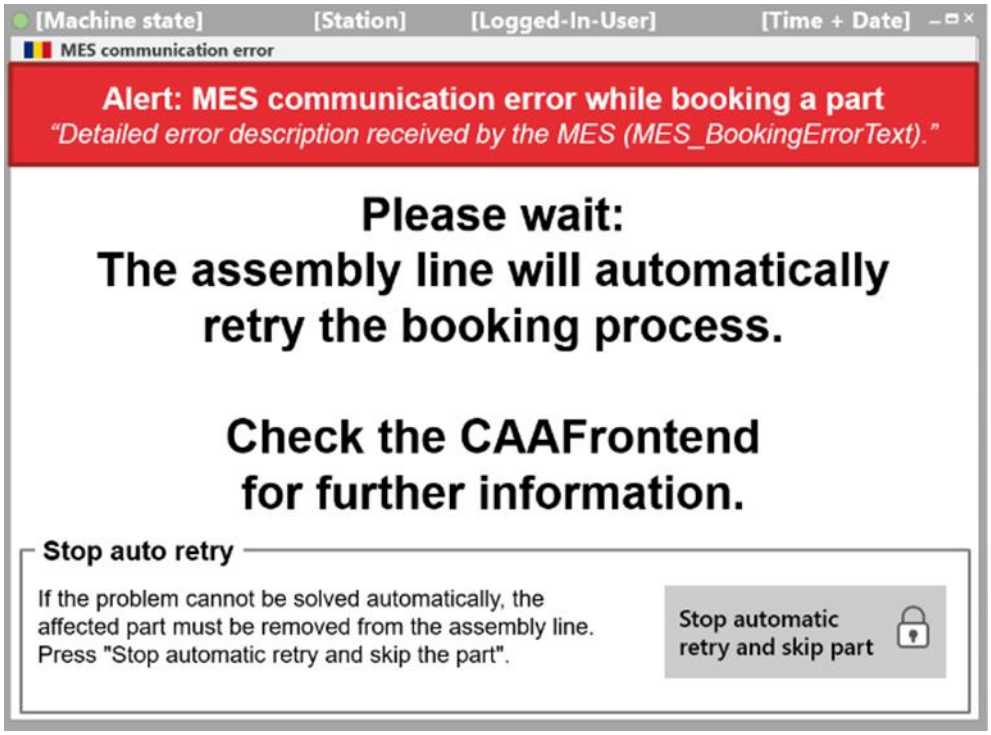


Figure 29: Machine HMI example - an unexpected communication error occurred

At the same time MES will start an automated retry logic since some of those problems can be fixed without any user interaction on machine side.

Note

- The automatic MES retry logic can be activated and deactivated by DRÄXLMAIER.
- By default, this logic should be enabled.
- Get in contact with the DRÄXLMAIER Group to clarify whether the retry logic is enabled.

If the problem can be fixed automatically by MES, MES will set `MES_BookingError=0` as soon as the error has been resolved. With this trigger, the assembly line should inform the operator that the error has been fixed and that the process can be restarted.

The following figure shows an example of such a machine HMI page:

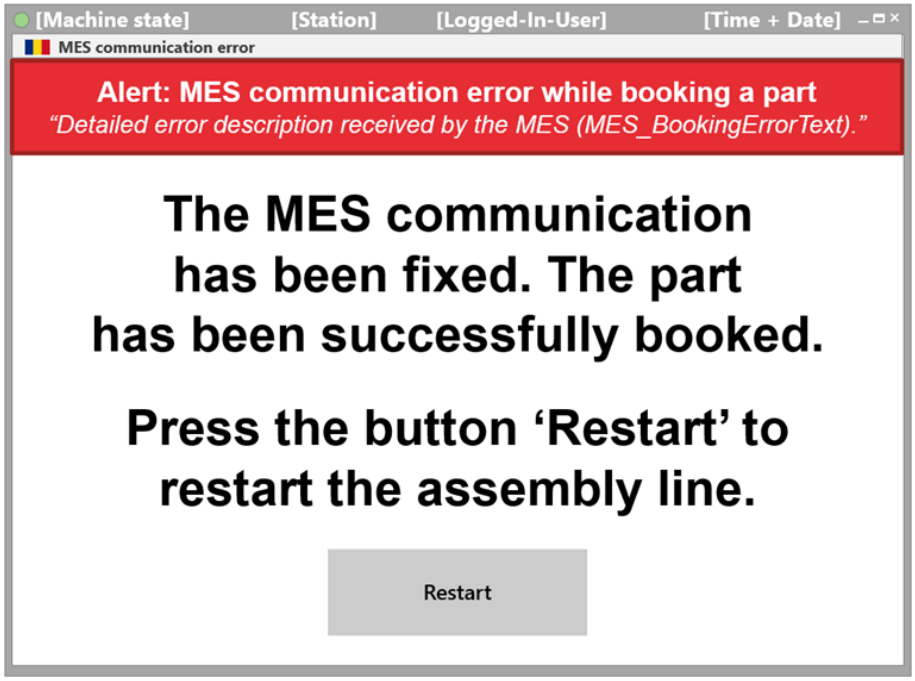


Figure 30: Machine HMI example - the communication error has been fixed automatically

After the "Restart" button has been pressed the assembly line continues with the regular routine by waiting for `MES_DataProcessed=1`.

Note

- For highly automated assembly lines the "Restart" button can be omitted.
- Thus, no user interaction is required.
- Get in contact with the DRÄXLMAIER Group to clarify if the "Restart" button must be used.

If the error cannot be fixed by MES automatically the operator has to press the “Stop automatic retry and skip part” button on the machine HMI:

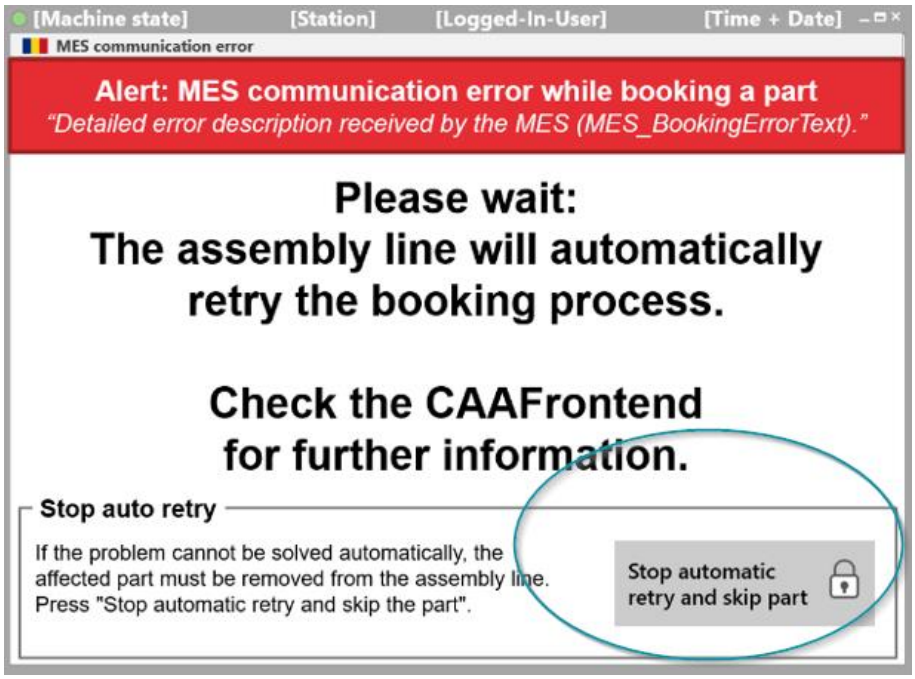


Figure 31: The button "Stop automatic retry and skip part"

After pressing the button “Stop automatic retry and skip part" the assembly line should inform the worker, that the affected part must be placed in the NOK box.

The following figure shows an example of such a machine HMI page:

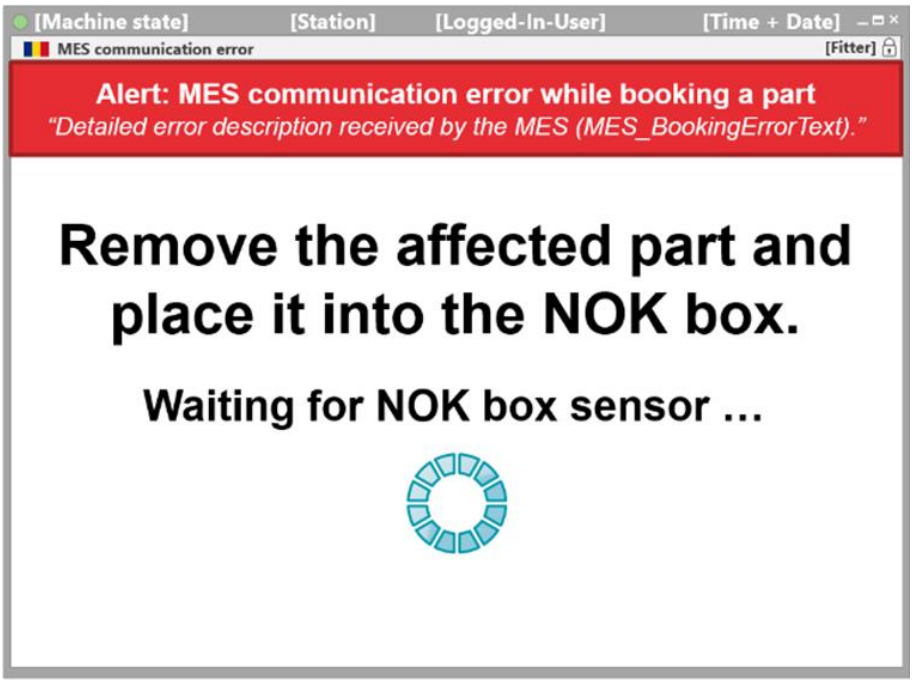


Figure 32: Machine HMI example - the affected part must be placed in the NOK box

Once the affected part has been detected in the NOK box, the assembly line should inform the worker.

The following figure shows an example of such a machine HMI page:



Figure 33: Machine HMI example - the NOK part has been detected, the problem is solved

As soon as the button "Restart" has been pressed, the assembly line must clear all existing process data for this part and set `DataAvailable=0`. Also the part has to be reordered at machine automatically (increase the amount by 1).
Afterwards the assembly line continues with sending data for the next part.

Note

- During this workflow the active order remains untouched.
- The active order will not be removed by MES.

The following workflow shows the whole Interrupt Process again in a process diagram:

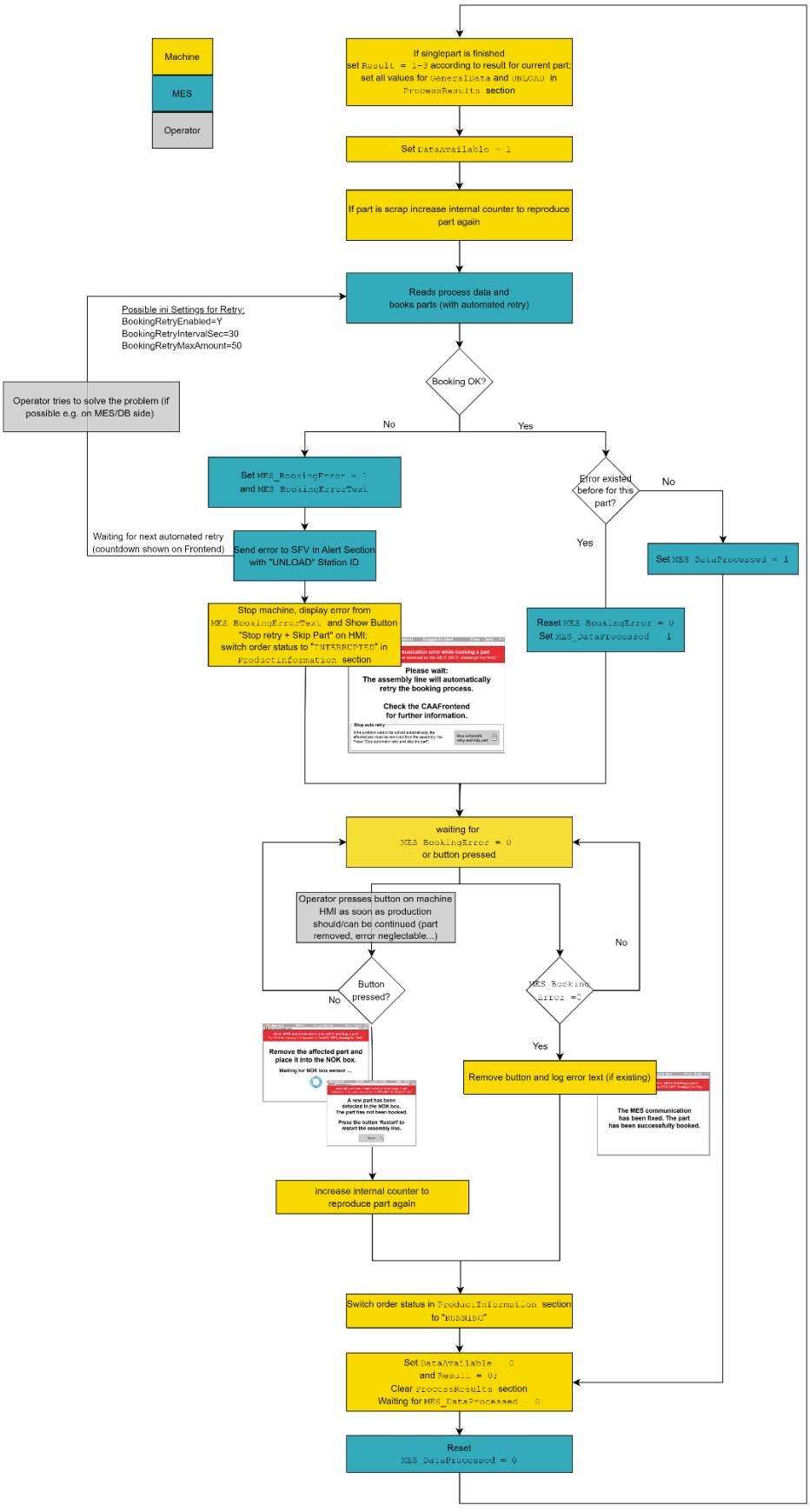


Figure 34: Workflow “Interrupt Process”

6.3 Concept 2: One-Piece-Flow Station

This concept must be used whenever only one part is inside the machine. Here the part is put into the machine, machine starts working on the part and at the end the part is removed from machine. After this the machine is waiting for the next part to be in place.

When this concept is in use, only the following WORKPLACE_LOAD_UNLOAD has to be used instead of the combination of WORKPLACE_LOAD and WORKPLACE_UNLOAD.

6.3.1 WORKPLACE_LOAD_UNLOAD

The following workflows are needed to start an order and to unload a part on the assembly line.

Note

- Without further notice all workflows are based on the Standard Handshake which can be found in the applicable document InterfaceSpecification&Structure_OPC_UA_template_V3_xx.pdf.

6.3.1.1 Workflow “Start Order“

To start an order on the machine the MES is starting a process with the `MES_Action "START"` using the standard Handshake.

6.3.1.2 Workflow “Provide Result Data“

As soon as a part has been finished at the machine, the result (including the process data) must be provided to MES.

Important

- It must be assured that the data is completely written to the OPC UA server (all parameters are up to date) before the CAAFrontend is informed by setting `DataAvailable` to 1.
- It must be assured that all nodes belonging to the OPC structure `ProcessResults` have been cleared and initialized by the PLC before new process data is written to it. See the chapter “8 Initializing process- and traceability data”.

7 Functional Requirements

7.1 Integrated Scanners

In case there are scanners integrated in the assembly line (auto scanner) it must be differentiated between scanning the main part or checking the commissioned part:

⇒ **Scanning the main part**
See chapter “Auto Scan to start order” in document “InterfaceSpecification&Structure_OPC_UA_template_V3_xx.pdf”

⇒ **Scanning a commissioned part**
See chapter “Auto scan inside process” in document “InterfaceSpecification&Structure_OPC_UA_template_V3_xx.pdf”

Note

- The final scanner configuration is developed within the project.

8 Initializing process- and traceability data

Process- and traceability data from a pre-process must be prevented from being incorrectly assigned to a new part. For this reason, the process- and traceability data must be initialized before starting the production of a new part.

These are the steps to initialize process- and traceability data:

- Process- and traceability data points are initialized during the “Provide Result Data” Workflow.
- After MES has confirmed the reading of the process- and traceability data points from the last part (with `MES_DataProcessed = 1`), the process- and traceability data points for the new part must be initialized.
- Each data point within the OPC structure `Workplace.ProcessResults` must be initialized with a default value.
- Default values are OPC UA data type dependent.
- The default values can be found in the following document: “InterfaceSpecification&Structure_OPC_UA_template_V3_03.pdf (ff).”

9 Reporting of process data for semifinished parts

By default, CS + HV assembly lines report process data only for finished parts (ZSBs) at the end of the manufacturing process:

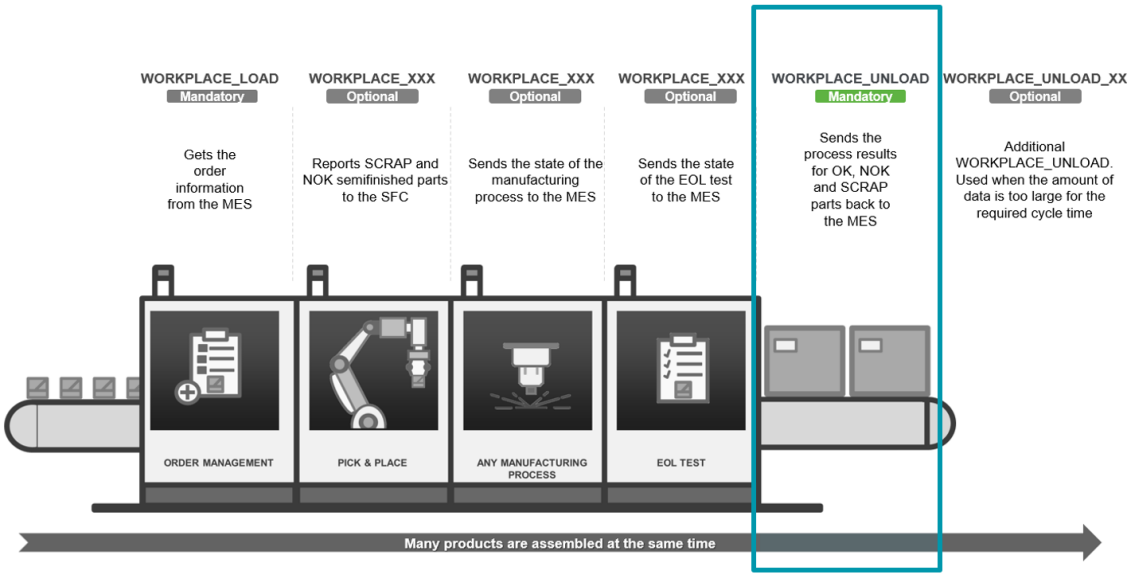


Figure 35: Process data reporting for finished parts (ZSBs) (by default)

In some cases, however, it is necessary to also report process data of semifinished parts directly at the workplace where they occur, e.g. to track and evaluate NOK or SCRAP parts in a more detailed way:

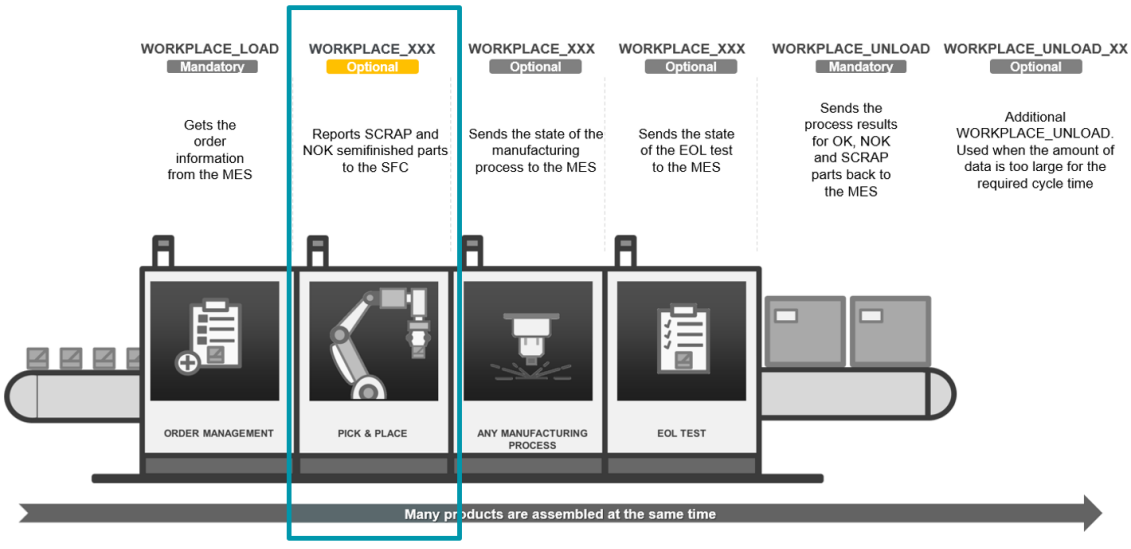


Figure 36: Process data reporting for semifinished parts (if requested)

Examples:

- Reporting process data of PCBAs
- Reporting process data of busbars

Contact the DRÄXLMAIER Group to clarify whether the reporting SCRAP and NOK semifinished parts is required for your assembly line.

10 Format of an error code for NOK and SCRAP components

If the result of the production process is NOK or SCRAP (`Result = 2`), the cause of the problem must be reported within the `ProcessResults` section in the OPC-UA process values `ErrorcodesNOK`, `ErrormessagesNOK`, `ErrorcodesSCRAP` and `ErrormessagesSCRAP`, depending on the final status of the part.

The detailed format is described in the following chapters.

Important

- The content may not contain special characters that are also used within an XML notation, e.g. &, <, >

10.1 Error codes at EOL, HV or Runin workplaces

Note

- This format of error codes is only valid for EOL-,HV- and RUN-In Test Stations. For all other stations the definition can be found in chapter “10.2 - Error codes at other workplaces”.
- Get in contact with the DRÄXLMAIER Group to clarify whether the station is working with these error codes.

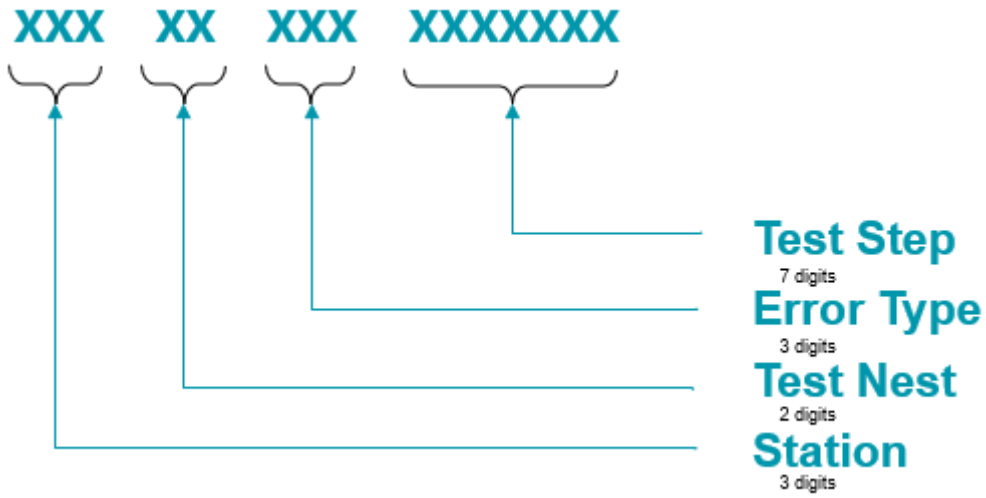


Figure 37: Format of an error number at EOL/HV specific workplaces

Example:

EOL02CUR4875546
RUN05OHM6876645

EOL	-	The error happened in station EOL.
02	-	The error happened in nest 02 of the EOL station.
CUR	-	The error was something with current
4875546	-	The error occurred in Test Step 4875546.

10.1.1 Station ID

The ID of the desired station defined by the layout.

- 3 digits uppercase abbreviation.

Examples:

- `EOL` EOL Station
- `HVT` HV Test Station
- `RUN` Runin Station

10.1.2 Nest ID

Defines multiple available nests in a process

- Only numeric double-digits with leading `0`.
- If the process uses multiple nests start to count at `01` (00 is reserved).
- If the process uses only one single nest `00` is used.

10.1.3 Error Type

Defines the type of the error.

- 3 digits uppercase abbreviation

Error Types with Test Step:

- `CUR` Current
- `RES` Resistance
- `VOL` Voltage
- `PRD` Product
- `VAL` Value
- `COM` Communication
- `TMP` Temperature

Error Types without Test Step: (Example: EOL02CON0000000)

- `CON` Contacting
- `GEN` General error (In the case of a general error, the error text must be described)

10.1.4 Test Step

Defines the number of the test step inside the testing process

- Amount of digits depending on test step length

10.2 Error codes at other workplaces

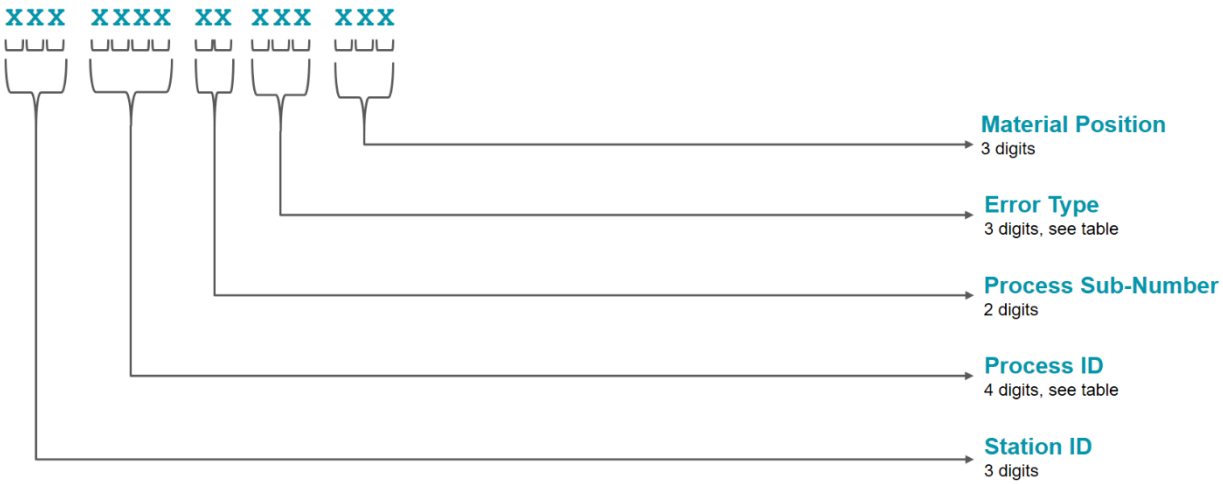


Figure 38: Format of an error number at other workplaces

Example:

M03OCRD02015002

M03	-	The error happened in station M03.
OCRD	-	The process was a camera OCR detection (e. g. reading the rating of a fuse).
02	-	The error occurred on the second camera.
015	-	The error type is “015 -> OCR detection NIO”.
002	-	The fuse in position 2 was affected.

10.2.1 Station ID

The ID of the desired station defined by the layout.

- 3 digits.
- Missing digits will be filled with leading `0` characters.

Examples**:**

- M03
- 180

10.2.2 Process ID

The technology of the desired station.

- 4 digits uppercase abbreviation.
- The abbreviations are predefined by the DRÄXLMIER Group.

Examples:

- `TOXE` TOX®-Round Joint sheet metal joining procedure
- `AIRF` Airflow test
- `LAWE` Laser welding
- `USWM` Ultrasonic welding membrane
- `OCRD` Camera OCR detection
- `LAMA` Laser marking

- `SCRE` Screwing
- `HEST` Heat staking
- `COCH` Connector check
- `MNIO` Manual NIO
- `ASPA` Assembly part
- `CLCH` Clip check
- `PRES` Press
- `VISO` Vision sensor
- `CHKL` Checklist
- `CMEA` Camera measurement
- `CRCO` Camera read code
- `CONC` Continuity check
- `FITT` Fitting with monitoring
- `LEAK` Leakage test
- `USWP` Ultrasonic welding plastic
- `USST` Ultrasonic stacking

Note

- Get in contact with the DÄXLMAIER Group to get a complete list with process IDs.

10.2.3 Process Sub-Number

Defines multiple available components in a process (e.g. more cameras in a station, more TOX tongs …).

- Only numeric double-digits with leading `0`.
- If the process uses multiple components start to count at `01` (00 is reserved).
- If the process uses only one single component `00` is used.

10.2.4 Error Type

Defines the type of the error.

- Three numeric digits according to a predefined list of the DRÄXLMAIER Group.

Examples:

`001` Length too short
`002` Length too long
`003` Position NIO
`004` Not available
`005` Temperature too high
`006` Temperature too low
`007` Pressure too high
`008` Pressure too low
`009` Airflow too high
`010` Airflow too low
`011` Force too high
`012` Force too low
`013` Offset not reached
`014` Offset exceeded
`015` OCR detection NIO
`016` Press force too high
`017` Press force too low
`018` Press way too high
`019` Press way too low
`020` Screw torque too high
`021` Screw torque too low

Note

- Get in contact with the DRÄXLMAIER Group to get the complete list with error types.

10.2.5 Material Position

Defines the position of the faulty material (e.g. the position of a fuse on a part).

- A three-digit number according to the “MES parameter list”.
- Leading `0` characters are used.
- If the position of the material has no meaning, `000` is used.

Note

- The MES parameter list defines the position of the materials on the part. Get in contact with the DRÄXLMAIER Group to get the latest MES parameter list.

11 FAQs

11.1 In which section of the information model do I find the process- and traceability data points?

The process- and traceability data points are in the following section:

Workplace.ProcessResults

11.2 How are OPC UA process values named?

In the past, the names of the OPC UA process values could be selected by the manufacturer. This is no longer possible because the names of the OPC UA process values have been standardized. The names of the process values are clearly defined by the OPC UA Nodeset.

11.3 How is the ProductID composed?

⇒ **See the chapter “MES Input parameters” in the document “InterfaceSpecification&Structure_OPC_UA_template_V3_xx.pdf”.**

11.4 Do we need to support NTP (Network Time Protocol)?

The Network Time Protocol is not supported in the network of the DRÄXLMAIER Group. The synchronization of the time will be handled by using the OPC UA Tag `MES_SyncDateTime`.

12 Glossary

Term	Description
OPC UA	Abbreviation for “Open Platform Communications Unified Architecture”.
HV	Abbreviation for “High Voltage”. A department of the DRÄXLMAIER Group.
CS	Abbreviation for “Component Systems”. A department of the DRÄXLMAIER Group.
MES	Abbreviation for “DRÄXLMAIER Group MES-System”.
Emergency Mode	The assembly line should provide an emergency mode in the case that the connection to the MES has been lost. In this case the assembly line should offer the ability to start an order without support of the MES. This feature is intended for use only in critical emergency cases.
Integrated or auto scanner	If a scanner is controlled by the assembly line (not by the MES), the scanner is called auto scanner or machine integrated scanner.
CAAFrontend	A Software from the DRÄXLMAIER Group. It is installed on a Panel PC directly at the assembly line and connects the assembly line with the MES. Using this software the worker at the assembly line can start an order, book OK parts and scrap NOK parts.
DRÄXLMAIER Shopfloor Systems	Shopfloor Integration Bus, MES, Shopfloor Visualization, Product Data Visualization
BOM	Bill of materials, which lists all the components for the assembling process.

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Folder / Name	Type	Description	Write Access	UNIT	Example
Root					
Objects					
DRX					
Globaldata	Struct				
Metric	Struct	The metric section is used to monitor the status of a CPU. For Siemens TIA there are standard functions available to get the values for some of these variables. ⚠ • If the machine is not capable to report any of these values the DRÄXLMAIER Group has to be informed as soon as possible.			
PLCSystemInfo	Variable	metrics of plc			
ChecksumStandardPLCProg	String[18]	checksum of standard blocks	Station		
ComLoadLastCycle	UInt16	% Communication load last cycle	Station		
CurrentCycleTimePLCProg	Int64	Current cycle time of plc program in ms	Station		
LongestCycleTimePLCProg	Int64	Longest cycle time of plc program in ms	Station		
MaxCycleTimePLCProg	Int64	max allowed cycle time of plc program in ms	Station		
ShortestCycleTimePLCProg	Int64	Shortest cycle time of plc program in ms	Station		
Tracing	Struct	Tracing data is supposed to show the overall status of the machine. All values should be shown and changed realtime. Data will be collected in fixed intervals and not within the productflow. ⚠ • If the machine is not capable to report any of these values the DRÄXLMAIER Group has to be informed as soon as possible.			
AlertActive	Bool	Flag to signalize if an error is active (global level)	Station		true/false
DateTime	DateTime (Siemens: LDT)	Current Date and Time on PLC. Affected by: `MES_SyncDateTime` Date time formatted like : yyyy-MM-dd HH:mm:ss	Station		2019-02-20 19:15:56
EquipmentExemplarNumber	String[18]	A static value, provided by the DRÄXLMAIER Group. The required value can be taken from one of the following DRÄXLMAIER systems: - PLM: Ressource number (starting with 5000) - Host: Optimain number 3 digits (added to Main Number) - SAP: 18 digits - only 10 used (unique when combined with Main Number)	Station		PLM: 5000000823-001 Optimain: 001
EquipmentMainNumber	String[13]	A static value, provided by the DRÄXLMAIER Group. The required value can be taken from one of the following DRÄXLMAIER systems: - PLM: Construction type (starting with 3000) - Host: Optimain number 5 digits - SAP: 13 digits DRÄXLMAIER material number ZFHM	Station		PLM: 300010752-001 Optimain: 87245
InterfaceVersion	String[5]	Version of InterfaceDescription provided by DRX, to be filled by machine manufacturer (according to state of implemetation)	Station		1.00
ResetDate	DateTime (Siemens: LDT)	Date when "total" values have last been reset Date time formatted like : yyyy-MM-dd HH:mm:ss	Station		2019-02-12 12:23:20
SoftwareChecksum	Array[0..x] of String[50]	Hash Checksum of each PLC ⚠ • For normal CPUS using Siemens TIA the checksum can be calculated with GetChecksum function. • For F-CPUS the F-Programsignature should be used	Station		26 5F 6B 9A 5B 01 1D 34
SoftwareVersion	String[20]	The version of the machine software. The format of the version is not fixed. ⚠ • This is not the version of the used Siemens TIA portal! This is the version of the machine software created with Siemens TIA portal.	Station		2017-10-3
TotalMaintenanceTime	Int32	Total time in maintenance (hours) (from reset date)	Station	h	5
TotalNOKParts	Int32	Total amount of NOK parts produced on machine (from reset date); Sum of all stations	Station	pcs	10
TotalOKParts	Int32	Total amount of OK parts produced on machine (from reset date); Sum of all stations	Station	pcs	10
TotalProducedParts	Int32	Total amount of parts produced on machine (from reset date); Sum of all stations	Station	pcs	300
TotalRunningTime	Int32	Total running hours (hours with power supply)	Station	h	2350
TotalScrapParts	Int32	Total amount of scrapped parts produced on machine (from reset date); Sum of all stations	Station	pcs	10
Energy	Struct	Container for all energy monitoring data, dependent on the sensors built in in the machine			
Electricity	Array [0..x] of Struct				
MeterId	String[20]	Identifier of the meter, to enable logical assignment to a physical substation (station ID)	Station		
ActiveEnergy	Float	Total consumed energy since reset	Station	Wh	
ActivePower	Float	Current consumed energy	Station	W	
ApparentPower	Float		Station	VA	
ReactiveEnergy	Float		Station	varh	
ReactivePower	Float		Station	var	
PowerFactor	Float	CosPhi	Station		
CompressedAir	Array [0..x] of Struct				
MeterId	String[20]	Identifier of the meter, to enable logical assignment to a physical substation (station ID)	Station		
Pressure	Float		Station	bar	
FlowRate	Float		Station	Nm³/h	
Count	Float		Station	Nm³	
Gas	Array [0..x] of Struct				
MeterId	String[20]	Identifier of the meter, to enable logical assignment to a physical substation (station ID)	Station		
FlowRate	Float		Station	Nm³/h	
Count	Float		Station	Nm³	
Cooling	Array [0..x] of Struct				
MeterId	String[20]	Identifier of the meter, to enable logical assignment to a physical substation (station ID)	Station		
ActiveEnergy	Float		Station	Wh	
ActivePower	Float		Station	W	
Heat	Array [0..x] of Struct				
MeterId	String[20]	Identifier of the meter, to enable logical assignment to a physical substation (station ID)	Station		
ActiveEnergy	Float		Station	Wh	
ActivePower	Float		Station	W	
Water	Array [0..x] of Struct				
MeterId	String[20]	Identifier of the meter, to enable logical assignment to a physical substation (station ID)	Station		
FlowRate	Float		Station	m³/h	
Count	Float		Station	m³	
Environment	Array [0..x] of Struct				
MeterId	String[20]	Identifier of the meter, to enable logical assignment to a physical substation (station ID)	Station		
LightIntensity	Float		Station	lux	
Temperature	Float		Station	°C	
Humidity	Float		Station	%	
OperatingHours	Float		Station	h	
Alerts	Struct	Structure for transferring detailed information about alerts			
FunctionalGroups	Array [0..x] of Struct	Struct for each functional group in station			
FunctionalGroupName	String[20]	Name of functional group	Station		Workplace_XX, Roboter_01
AlertActive	Bool	Flag to signalize if an alert is active (on functional group level)	Station		true/false
AlertDetails_Count	Int16	Amount of actually filled elements in AlertDetails array	Station		10
AlertDetails	Array [0..x] of Struct	Details for every alert on machine			
AlertCode	Int32	Code of an active alert, to be defined by manufacturer.	Station		[45,23,0,0]
AlertLevel	Int16	Severance of the alert: 1 = ERROR 2 = WARNING 3 = INFO ⚠ • The moment for setting an Alert with AlertLevel = ERROR is not fixed but depends on the machine and must be discussed with the supplier. • Setting an alert with AlertLevel = ERROR will cause the Shopfloor Visualization to display the machine state as ERROR. In this case, all machine states sent from the line within the variables `Idle`, `Running` etc. are not taken into account for displaying. • Setting an alert with AlertLevel = WARNING or AlertLevel = INFO will not cause the Shopfloor Visualization to change the displayed machine state to ERROR. In this case, the displayed machine state in the Shopfloor Visualization depends on the variables `Idle`, `Running` etc.. **Example:** An alert with AlertLevel = ERROR is set at the moment an error on the line becomes active and no further components are produced. In this case the Shopfloor Visualization will display a red ERROR machine state.	Station		[1,2,0,0]
AlertMessage	String[100]	Textual description of the active alert. Matching the code in AlertCode.	Station		["Printer is not responding (Check power/connection)", "Roundtable Drive Profinet Connection Error",.]
WORKPLACE_LOAD	Struct				
Info	Struct	Important Information for current station/workplace			
MES_StationID	String[20]	Station ID in order to identify the current station	MES		STATION_WELDING_01 ST0110
MES_SyncDateTime	DateTime (Siemens: LDT)	Date time to sync with plc Date time formatted like : yyyy-MM-dd HH:mm:ss	MES		2019-02-20 19:15:56
MES_Workcenter	String[50]	MES Workcenter which is used at the current station	MES		DI345678 ZUFUEHRUNG
Status	Struct	Status data of current workplace. The status data is shown in the Shopfloor Visualization as the machine state. ⚠ • If the machine is not capable to report any of these values the DRÄXLMAIER Group has to be informed as soon as possible.			
Active	Bool	False: station/turntable/control side deactivated or not available True: station/turntable/control side activated	Station		true/false
AutomaticMode	Bool	False: Plc in manual mode, not controlled by mes system True: Plc controlled by mes system	Station		true/false
Equipping	Int16	0: Machine is currently not equipping 1: Machine is beeing equipped with new tools -1: If the machine is not capable to report the equipping mode.	Station		0
Idle	Bool	False: PLC not ready for MES_Release. PLC is not ready to start production (is running, in maintenance,...) True: PLC ready for next part to produce ⚠ • If the production line supports only one machine state (one state for the whole production line) make sure to use the same state in WORKPLACE_UNLOAD.Status.Idle.	Station		true/false
LastCycleTime	Float	Cycle time for the current part from start to end of production (machine time + human time) Measurement start: 1st activity which can be detected by machine Measurement end: last activity which can be detected by machine	Station		23.40
MaintenanceMode	Int16	0: Machine is currently running on normal mode 1: Machine is running in maintenance mode -1: If the machine is not capable to report the maintenance mode.	Station		1

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Running	Bool	Affected by: MES_Release False: PLC production not in progress True: Production in progress, part is processed	Station		true/false
Sampling	Int16	0: machine is currently not in sampling mode 1: machine is running test/teaching parts -1: machine is not capable to report the sampling mode since it is not available.	Station		-1
TotalNOKParts	Int32	Number of NOK parts on this specific workplace. ⚠ • NOK parts can be reworked.	Station		0
TotalOKParts	Int32	Number of OK parts on this specific workplace	Station		1
TotalScrapParts	Int32	Number of SCRAP on this specific workplace ⚠ • SCRAP parts can not be reworked.	Station		0
ProductInformation	Array [0..x] of Struct	Information about all active products in current workplace. ⚠ • This information is mandatory also if the cycle time is quite low.			
ProductID	String[100]	Currently processed ProductID (based on MES_ProductID).	Station		BM2F094126CA3320; 300000220-001P001007827
ProductType	String[50]	Currently processed Product Type (based on MES_ProductType).	Station		BM2F094112;300000220-001
Status	Int16	0: not available 1: WAITING 2: RUNNING 3: CANCELLED 4: INTERRUPTED	Station		1
Workflow	Struct	to be used if any actions are done between MES and current station			
MES	Struct	all values written by MES			
MES_Action	String[20]	START, CANCEL, REORDER	MES		SCAN, WELD, CANCEL,CRIMP,..
MES_CancelProcess	Bool	False: Process not canceled True: Process canceled due to any error ⚠ • This value is typically not used by production lines in the segment CS (if they are using only to 2 workplaces).	MES		true/false
MES_DataProcessed	Bool	Signals if result has been read and processed Affected by: DataAvailable False: MES system has not processed data True: MES system has read processdata. DataAvailable must be set to 0	MES		true/false
MES_PartsToProduce	Int32	Total amount of parts to be produced for the current job ⚠ • Can count up to 999999	MES		e.g. 234897 or e.g. 21
MES_ProductID	String[100]	The ProductID of the requested job. Used to identify the requested product for this new job. Used to link the process data to this new job. ⚠ • See the chapter "ProductID" for a detailled explanation of the used format.	MES		300000220-001P001007827000001; BM2F094126CA3320; E9904969P40102
MES_ProductType	String[50]	The ProductType of the requested job.	MES		300000220-001; BM2F094112; P401
MES_Release	Bool	production release False: production not allowed, don't start machine True: start production	MES		true/false
MES_ProductParameter	Struct	These variables are usually added when additional input parameters from MES to the machine are required.			
MES_CustomerChangeIndex	String[6]	Customer change index ⚠ • The format of the customer change index (from MES field "ordInfoField3") is not fixed. • Get in contact with the DRÄXLMAIER group to get the latest example.	MES		01S
MES_CustomerChangeLevel	String[15]	Customer change level ⚠ • The format of the customer change level (from MES field "ordInfoField2") is not fixed. • Get in contact with the DRÄXLMAIER group to get the latest example.	MES		22.10.2015
MES_CustomerNumber	String[20]	Customer number ⚠ • The format of the customer number (from MES field "ordInfoField1") is not fixed. • Get in contact with the DRÄXLMAIER group to get the latest example.	MES		7641798-03
MES_JobNr	Int16	Can be used to define different product variants that have to be handled on machine side 0: not available (To be defined)	MES		1
MES_PackagingType	String[15]	Box ID (Behälter ID from MES)	MES		APE1
MES_StartSinglePartID	Int32	Singlepartnumber for starting the numeration on the machine. At the first start usually "1", after restarting an order this number will contain the last number from the result data incremented by 1. ⚠ • Can count up to 999999	MES		992330
MES_MaterialList	Array[0..x] of Struct	List of needed materials. The format of the list is project specific. Please get in contact with DRX to get a	MES		
MES_MaterialNr	String[100]	Material number of current material	MES		300000220-001; T0101589
MES_BoxNr	Int16	Box number of current material	MES		2
MES_Handworkplace	Int16	Number of the desired hand-workplace fur current material	MES		5
MES_ToBeDefined		Placeholder for other additional variables. ⚠ • These variables are optional and not mandatory. • These additional variables must be agreed between DRÄXLMAIER and the manufacturer.	MES		e.g. FazitString
Station	Struct	all values written by PLC			
DataAvailable	Bool	False: data not available or not related to current production process True: data available and actual. Must not be set to 0 until MES_DataProcessed is set to 1	Station		true/false
Result	Int16	Result of loading process 0: not available 1: OK / INPRODUCTION 2: NOK 3: CANCEL	Station		2
ControlParameter	Struct	Placeholder for needed values which are sent by the machine to the MES			
ToBeDefined		Placeholder for other additional variables. ⚠ • These variables are optional and not mandatory. • These additional variables must be agreed between DRÄXLMAIER and the manufacturer.	Station		
ProcessResults	Struct	all values written by PLC to specify the process result. The process variables have to be defined together with DRX at the start of the project. Datatypes can be defined by manufacturer according to needed requirements of content.			
GeneralData					
ErrorcodesNOK	Array [0..x] of String[30]	Error codes for each part that was sent as NOK from machine. ⚠ • The error codes have to be defined by the machine manufacturer. • The error codes have to use the format specified in chapter "Format of an Error Number".	Station		[2,0,0,0]
ErrorcodesScrap	Array [0..x] of String[30]	Error codes for each part that was sent as Scrap from machine ⚠ • The error codes have to be defined by the machine manufacturer. • The error codes have to use the format specified in chapter "Format of an Error Number".	Station		[4,0,0,0]
ErrormessagesNOK	Array [0..x] of String[150]	Textual description of an error code from field ErrorcodesNOK ⚠ • One message for each error code in ErrorcodesNOK • Errormessage(s) should be collected for each part and have to be deleted after each cycle (MES_DataProcessed=1) • The content may not contain special characters that are also used within an XML notation, e.g. &, <, >	Station		[Torque out of limit,,]
ErrormessagesScrap	Array [0..x] of String[150]	Textual description of an error code from field ErrorcodesScrap ⚠ • One message for each error code in ErrorcodesScrap • Errormessage(s) should be collected for each part and have to be deleted after each cycle (MES_DataProcessed=1) • The content may not contain special characters that are also used within an XML notation, e.g. &, <, >	Station		[Surface too hot,,,]
PartsNOK	Int32	Number of NOK parts for current process. ⚠ • NOK parts can be reworked.	Station		0
PartsOK	Int32	Number of OK parts for current process.	Station		1
PartsScrap	Int32	Number of SCRAP parts for current process.. ⚠ • SCRAP parts can not be reworked.	Station		0
ProductID	String[100]	Currently processed ProductID (based on MES_ProductID). ⚠ • See the chapter "ProductID" for a detailled explanation of the used format.	Station		BM2F094126CA3320; 300000220-001P001007827
ProductType	String[50]	Currently processed Product Type (based on MES_ProductType).	Station		BM2F094112;300000220-001
AlternativeID	Array [0..x] of String[100]	Barcode (e. g. DMC, Code128, ...) on the final part, that is/are different from the MES internal ProductID and that might be needed for proper tracking of process data in MES. Used to link the MES internal part data to the physical machine part data. ⚠ • The content of the AlternativeID is not fixed and is optional to be filled. • Get in contact with the DRÄXLMAIER group for a final definition if the filling is needed and which value should be filled in.	Station		BUC13092023111299233000000220-001P001007827
TotalProcessingTime	Float	Processing time (duration) for all involved actions in total	Station	minutes	1.87
NAME of ACTION		to be used for process specific data for current action			

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ToBeDefined		Placeholder for needed values	Station		
WORKPLACE_XX	Struct	Live data of station XX (if more stations are used within the machine). ⚠ • The use of more than the two WORKPLACES besides WORKPLACE_LOAD and WORKPLACE_UNLOAD is optional. • The use of additional WORKPLACES must be agreed with the DRÄXLMAIER Group. • Additional WORKPLACES are used for detailed monitoring of a specific process.			
Info	Struct	Important Information for current station/workplace			
MES_StationID	String[20]	Station ID in order to identify the current station	MES		STATION_WELDING_01 ST0110
MES_SyncDateTime	DateTime (Siemens: LDT)	Date time to sync with plc Date time formatted like : yyyy-MM-dd HH:mm:ss	MES		2019-02-20 19:15:56
MES_Workcenter	String[50]	MES Workcenter which is used at the current station	MES		DI345678 ZUFUEHRUNG
Status	Struct	Status data of current workplace. The status data is shown in the Shopfloor Visualization as the machine state. ⚠ • If the machine is not capable to report any of these values the DRÄXLMAIER Group has to be informed as soon as possible.			
Active	Bool	False: station/turntable/control side deactivated or not available True: station/turntable/control side activated	Station		true/false
AutomaticMode	Bool	False: Plc in manual mode, not controlled by mes system True: Plc controlled by mes system	Station		true/false
Equipping	Int16	0: Machine is currently not equipping 1: Machine is beeing equipped with new tools -1: If the machine is not capable to report the equipping mode.	Station		0
Idle	Bool	False: PLC not ready for MES_Release. PLC is not ready to start production (is running, in maintenance,...) True: PLC ready for next part to produce ⚠ • If the production line supports only one machine state (one state for the whole production line) make sure to use the same state in WORKPLACE_UNLOAD.Status.Idle.	Station		true/false
LastCycleTime	Float	Cycle time for the current part from start to end of production (machine time + human time) Measurement start: 1st activity which can be detected by machine Measurement end: last activity which can be detected by machine	Station		23.40
MaintenanceMode	Int16	0: Machine is currently running on normal mode 1: Machine is running in maintenance mode -1: If the machine is not capable to report the maintenance mode.	Station		1
Running	Bool	Affected by: `MES_Release` False: PLC production not in progress True: Production in progress, part is processed	Station		true/false
Sampling	Int16	0: machine is currently not in sampling mode 1: machine is running test/teaching parts -1: machine is not capable to report the sampling mode since it is not available.	Station		-1
TotalNOKParts	Int32	Number of NOK parts on this specific workplace. ⚠ • NOK parts can be reworked.	Station		0
TotalOKParts	Int32	Number of OK parts on this specific workplace	Station		1
TotalScrapParts	Int32	Number of SCRAP on this specific workplace ⚠ • SCRAP parts can not be reworked.	Station		0
ProductInformation	Array [0..x] of Struct	Information about all active products in current workplace. ⚠ • This information is mandatory also if the cycle time is quite low.			
ProductID	String[100]	Currently processed ProductID (based on `MES_ProductID`).	Station		BM2F094126CA3320; 300000220-001P001007827
ProductType	String[50]	Currently processed Product Type (based on `MES_ProductType`).	Station		BM2F094112;300000220-001
Status	Int16	0: not available 1: WAITING 2: RUNNING 3: CANCELLED 4: INTERRUPTED	Station		1
WORKPLACE_UNLOAD	Struct	live data of final station (if more stations are used within the machine)			
Info	Struct	Important Information for current station/workplace			
MES_StationID	String[20]	Station ID in order to identify the current station	MES		STATION_WELDING_01 ST0110
MES_SyncDateTime	DateTime (Siemens: LDT)	Date time to sync with plc Date time formatted like : yyyy-MM-dd HH:mm:ss	MES		2019-02-20 19:15:56
MES_Workcenter	String[50]	MES Workcenter which is used at the current station	MES		DI345678 ZUFUEHRUNG
Status	Struct	Status data of current workplace. The status data is shown in the Shopfloor Visualization as the machine state. ⚠ • If the machine is not capable to report any of these values the DRÄXLMAIER Group has to be informed as soon as possible.			
Active	Bool	False: station/turntable/control side deactivated or not available True: station/turntable/control side activated	Station		true/false
AutomaticMode	Bool	False: Plc in manual mode, not controlled by mes system True: Plc controlled by mes system	Station		true/false
Equipping	Int16	0: Machine is currently not equipping 1: Machine is beeing equipped with new tools -1: If the machine is not capable to report the equipping mode.	Station		0
Idle	Bool	False: PLC not ready for MES_Release. PLC is not ready to start production (is running, in maintenance,...) True: PLC ready for next part to produce ⚠ • If the production line supports only one machine state (one state for the whole production line) make sure to use the same state in WORKPLACE_UNLOAD.Status.Idle.	Station		true/false
LastCycleTime	Float	Cycle time for the current part from start to end of production (machine time + human time) Measurement start: 1st activity which can be detected by machine Measurement end: last activity which can be detected by machine	Station		23.40
MaintenanceMode	Int16	0: Machine is currently running on normal mode 1: Machine is running in maintenance mode -1: If the machine is not capable to report the maintenance mode.	Station		1
Running	Bool	Affected by: `MES_Release` False: PLC production not in progress True: Production in progress, part is processed	Station		true/false
Sampling	Int16	0: machine is currently not in sampling mode 1: machine is running test/teaching parts -1: machine is not capable to report the sampling mode since it is not available.	Station		-1
TotalNOKParts	Int32	Number of NOK parts on this specific workplace. ⚠ • NOK parts can be reworked.	Station		0
TotalOKParts	Int32	Number of OK parts on this specific workplace	Station		1
TotalScrapParts	Int32	Number of SCRAP on this specific workplace ⚠ • SCRAP parts can not be reworked.	Station		0
ProductInformation	Array [0..x] of Struct	Information about all active products in current workplace. ⚠ • This information is mandatory also if the cycle time is quite low.			
ProductID	String[100]	Currently processed ProductID (based on `MES_ProductID`).	Station		BM2F094126CA3320; 300000220-001P001007827
ProductType	String[50]	Currently processed Product Type (based on `MES_ProductType`).	Station		BM2F094112;300000220-001
Status	Int16	0: not available 1: WAITING 2: RUNNING 3: CANCELLED 4: INTERRUPTED	Station		1
WorkflowUnload	Struct	to be used only if parts are unloaded at a different than the loading station and no other actions will take place			
MES	Struct	all values written by MES			
MES_BookingError	Bool	False: MES Booking is done without problems True: Booking error occurred on MES side. Process should be stopped!	MES		true/false
MES_BookingErrorText	String[150]	Contains the error text from MES that is shown on CaaFrontend screen if any error occurrs during booking.	MES		No more orders available
MES_DataProcessed	Bool	Signals if result has been read and processed Affected by: `DataAvailable` False: MES system has not processed data True: MES system has read processdata. `DataAvailable` must be set to 0	MES		true/false
Station	Struct	all values written by PLC			
DataAvailable	Bool	False: data not available or not related to current production process True: data available and actual. Must not be set to 0 until `MES_DataProcessed` is set to 1	Station		true/false
Result	Int16	Result of unloading process 0: not available 1: OK / INPRODUCTION 2: NOK 3: CANCEL	Station		1

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SecurityData	Struct	Placeholder for security relevant data which should not be visisble in the data structure of processresults. Structure should only be existing if it is in use for this machine/project ⚠ • These variables are optional and not mandatory. • These additional variables must be agreed between DRÄXLMAIER and the manufacturer.			
ProductID	String[100]	Currently processed ProductID (based on MES_ProductID).	Station		Format MES/HOST: BM2F094112CA3320 Format MES2/SAP: 300000220-001P001007827
CSR	String	CSR String of the product ⚠ The String length of this variable could expand a lot. Maximum String length will be defined within the project.	Station		
PublicKey	String	Public key of the product (base64 encoded DER) ⚠ The String length of this variable could expand a lot. Maximum String length will be defined within the project.	Station		
Signature	String	Signature of the product (base64 encoded) ⚠ The String length of this variable could expand a lot. Maximum String length will be defined within the project.	Station		
ToBeDefined		Placeholder for other additional variables. ⚠ • These variables are optional and not mandatory. • These additional variables must be agreed between DRÄXLMAIER and the manufacturer.	Station		
ProcessResults	Struct	all values written by PLC to specify the process result. The process variables have to be defined together with DRX at the start of the project. Datatypes can be defined by manufacturer according to needed requirements of content.			
GeneralData					
ErrorcodesNOK	Array [0..x] of String[30]	Error codes for each part that was sent as NOK from machine. ⚠ • The error codes have to be defined by the machine manufacturer. • The error codes have to use the format specified in chapter "Format of an Error Number".	Station		[2,0,0,0]
ErrorcodesScrap	Array [0..x] of String[30]	Error codes for each part that was sent as Scrap from machine ⚠ • The error codes have to be defined by the machine manufacturer. • The error codes have to use the format specified in chapter "Format of an Error Number".	Station		[4,0,0,0]
ErrormessagesNOK	Array [0..x] of String[150]	Textual description of an error code from field ErrorcodesNOK ⚠ • One message for each error code in `ErrorcodesNOK` • Errormessage(s) should be collected for each part and have to be deleted after each cycle (MES_DataProcessed=1) • The content may not contain special characters that are also used within an XML notation, e.g. &, <, >	Station		[Torque out of limit,,,]
ErrormessagesScrap	Array [0..x] of String[150]	Textual description of an error code from field ErrorcodesScrap ⚠ • One message for each error code in `ErrorcodesScrap` • Errormessage(s) should be collected for each part and have to be deleted after each cycle (MES_DataProcessed=1) • The content may not contain special characters that are also used within an XML notation, e.g. &, <, >	Station		[Surface too hot,,,]
PartsNOK	Int32	Number of NOK parts for current process. ⚠ • NOK parts can be reworked.	Station		0
PartsOK	Int32	Number of OK parts for current process.	Station		1
PartsScrap	Int32	Number of SCRAP parts for current process.. ⚠ • SCRAP parts can not be reworked.	Station		0
ProductID	String[100]	Currently processed ProductID (based on `MES_ProductID`). ⚠ • See the chapter "ProductID" for a detailled explanation of the used format.	Station		BM2F094126CA3320; 300000220-001P001007827
ProductType	String[50]	Currently processed Product Type (based on `MES_ProductType`).	Station		BM2F094112;300000220-001
AlternativeID	Array [0..x] of String[100]	Barcode (e. g. DMC, Code128, ...) on the final part, that is/are different from the MES internal ProductID and that might be needed for proper tracking of process data in MES. Used to link the MES internal part data to the physical machine part data. ⚠ • The content of the AlternativeID is not fixed and is optional to be filled. • Get in contact with the DRÄXLMAIER group for a final definition if the filling is needed and which value should be filled in.	Station		BUC13092023111299233000000220-001P001007827
TotalProcessingTime	Float	Processing time (duration) for all involved actions in total	Station	minutes	1.87
UNLOAD		to be used for specific process parameters that are not assigned to a special process			
FileName	String[50]	If additional data is transferred through Data2Cloud (via a file interface), this variable should contain the respective file name including the extension. To ensure uniqueness the file name should contain the ProductID plus a time stamp. ⚠ • If file transfer is expected, but the machine is unable to save the file in the designated folder, the machine has to return an ErrorMessage to signalize the problem to MES • If no file is to be saved, the value of this field should be empty	Station		A12345670101_20191205123001.zip
SinglePartID	String[6]	The part number of the currently processed single part. **Format:** Fixed length of 6 digits, use leading "0" characters if needed.	Station		000001
QualityPart	Bool	False: Part was produced normally True: Part was produced for QS reasons	Station		true/false
ToBeDefined		Placeholder for the process parameters of the finished part that don´t belong to a special process	Station		
ProcessTimes	Array of Struct	process times for each process			
EquipmentID	String[30]	ID of the source which reported this time span (e.g. StationID, ToolID,...)	Station		ST010_IR01
OperationID	String[20]	Operation which was performed during that time (e.g. pick, driving in, waiting,...)	Station		1.0020-10.10
Description	String[50]	short desciption	Station		
StartTime	DateTime (Siemens: LDT)	Start timestamp	Station		18.05.2020 13:00
EndTime	DateTime (Siemens: LDT)	End timestamp	Station		18.05.2020 13:15
MaterialConsumption	Struct	container for material consumption			
AdditionalConsumption_Count	Int16	Amount of actually filled elements in AdditionalConsumption array	Station		5
AdditionalConsumption	Array of Struct	list of all material commissioned additionally (exceeding the regular consumption)			
MaterialNr	String[100]	Host: Material label SAP VDA: Segment P SAP RML: Pos. 1-13	Station		
Consumption	Float	Consumed amount	Station		
Charge	String[30]	Host: Batch label SAP VDA: Supplier or DRÄXLMAIER internal Batch VDA Segment 1T SAP RML: Pos. 14-23	Station		
ExpirationDate	String[20]	optional SAP VDA: Segment 14D SAP RML: n/a	Station		
DeliveryNr	String[30]	optional SAP VDA: Segment 2S SAP RML: n/a	Station		
HandlingUnit	String[20]	SAP VDA: Segment 1J RML: Pos. 24-43	Station		
LabelContent	String[300]	whole 2D barcode label content for VDA including special characters has to be masked, e.g. **#&30;**	Station		
PickPosition	String[20]	optional SAP storage unit where the part was picked (hardcoded on PLC, eg. 0000-000-000-000)	Station		
PositionIndex	String[20]	Detailed position where the material was mounted in the product (optional, only needed if singlepart tracking is required)	Station		
SinglepartLabelSupplier	String[100]	Unique supplier label on the product for traceability. (optional, only needed if singlepart tracking is required)	Station		
RegularConsumption_Count	Int16	Amount of actually filled elements in RegularConsumption array	Station		5
RegularConsumption	Array of Struct	list of all commissioned material			
MaterialNr	String[100]	Host: Material label SAP VDA: Segment P SAP RML: Pos. 1-13	Station		
Consumption	Float	Consumed amount	Station		
Charge	String[30]	Host: Batch label SAP VDA: Supplier or DRÄXLMAIER internal Batch VDA Segment 1T SAP RML: Pos. 14-23	Station		
ExpirationDate	String[20]	optional SAP VDA: Segment 14D SAP RML: n/a	Station		
DeliveryNr	String[30]	optional SAP VDA: Segment 2S SAP RML: n/a	Station		
HandlingUnit	String[20]	SAP VDA: Segment 1J RML: Pos. 24-43	Station		
LabelContent	String[300]	whole 2D barcode label content for VDA including special characters has to be masked, e.g. **#&30;**	Station		
PickPosition	String[20]	optional SAP storage unit where the part was picked (hardcoded on PLC, eg. 0000-000-000-000)	Station		
PositionIndex	String[20]	Detailed position where the material was mounted in the product (optional, only needed if singlepart tracking is required)	Station		
SinglepartLabelSupplier	String[100]	Unique supplier label on the product for traceability. (optional, only needed if singlepart tracking is required)	Station		
ToBeDefined		Placeholder for the process parameters of the finished part belonging to the predefined processes for this station. The complete list of process parameters is provided by the DRÄXLMAIER group.	Station		
WORKPLACE_LOAD_UNLOAD / WORKPLACE_YY	Struct	structure for one-piece-flow station			
Info	Struct	Important Information for current station/workplace			

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MES_StationID	String[20]	Station ID in order to identify the current station	MES		STATION_WELDING_01 ST0110
MES_SyncDateTime	DateTime (Siemens: LDT)	Date time to sync with plc Date time formatted like : yyyy-MM-dd HH:mm:ss	MES		2019-02-20 19:15:56
MES_Workcenter	String[50]	MES Workcenter which is used at the current station	MES		DI345678 ZUFUEHRUNG
Status	Struct	Status data of current workplace. The status data is shown in the Shopfloor Visualization as the machine state. ⚠ • If the machine is not capable to report any of these values the DRÄXLMAIER Group has to be informed as soon as possible.			
Active	Bool	False: station/turntable/control side deactivated or not available True: station/turntable/control side activated	Station		true/false
AutomaticMode	Bool	False: Plc in manual mode, not controlled by mes system True: Plc controlled by mes system	Station		true/false
Equipping	Int16	0: Machine is currently not equipping 1: Machine is beeing equipped with new tools -1: If the machine is not capable to report the equipping mode.	Station		0
Idle	Bool	False: PLC not ready for MES_Release. PLC is not ready to start production (is running, in maintenance,...) True: PLC ready for next part to produce ⚠ • If the production line supports only one machine state (one state for the whole production line) make sure to use the same state in WORKPLACE_UNLOAD.Status.Idle.	Station		true/false
LastCycleTime	Float	Cycle time for the current part from start to end of production (machine time + human time) Measurement start: 1st activity which can be detected by machine Measurement end: last activity which can be detected by machine	Station		23.40
MaintenanceMode	Int16	0: Machine is currently running on normal mode 1: Machine is running in maintenance mode -1: If the machine is not capable to report the maintenance mode.	Station		1
Running	Bool	Affected by: `MES_Release` False: PLC production not in progress True: Production in progress, part is processed	Station		true/false
Sampling	Int16	0: machine is currently not in sampling mode 1: machine is running test/teaching parts -1: machine is not capable to report the sampling mode since it is not available.	Station		-1
TotalNOKParts	Int32	Number of NOK parts on this specific workplace. ⚠ • NOK parts can be reworked.	Station		0
TotalOKParts	Int32	Number of OK parts on this specific workplace	Station		1
TotalScrapParts	Int32	Number of SCRAP on this specific workplace ⚠ • SCRAP parts can not be reworked.	Station		0
ProductInformation	Array [0..x] of Struct	Information about all active products in current workplace. ⚠ • This information is mandatory also if the cycle time is quite low.			
ProductID	String[100]	Currently processed ProductID (based on `MES_ProductID`).	Station		BM2F094126CA3320; 300000220-001P001007827
ProductType	String[50]	Currently processed Product Type (based on `MES_ProductType`).	Station		BM2F094112;300000220-001
Status	Int16	0: not available 1: WAITING 2: RUNNING 3: CANCELLED 4: INTERRUPTED	Station		1
Workflow	Struct	to be used if any actions are done between MES and current station			
MES	Struct	all values written by MES			
MES_Action	String[20]	used workflow (To be defined)	MES		SCAN, WELD, CANCEL,CRIMP,..
MES_BookingError	Bool	False: MES Booking is done without problems True: Booking error occurred on MES side. Process should be stopped!	MES		true/false
MES_BookingErrorText	String	Contains the error text from MES that is shown on CaaFrontend screen if any error occurrs during booking.	MES		No more orders available
MES_CancelProcess	Bool	False: Process not canceled True: Process canceled due to any error ⚠ • This value is typically not used by production lines in the segment CS (if they are using only to 2 workplaces).	MES		true/false
MES_DataProcessed	Bool	Signals if result has been read and processed Affected by: `DataAvailable` False: MES system has not processed data True: MES system has read processdata. `DataAvailable` must be set to 0	MES		true/false
MES_PartsToProduce	Int32	Total amount of parts to be produced for the current job ⚠ • Can count up to 999999	MES		e.g. 234897 or e.g. 21
MES_ProductID	String[100]	The ProductID of the requested job. Used to identify the requested product for this new job. Used to link the process data to this new job. ⚠ • See the chapter "ProductID" for a detailled explanation of the used format.	MES		300000220-001P001007827000001; BM2F094126CA3320; E9904969P40102
MES_ProductType	String[50]	The ProductType of the requested job.	MES		300000220-001; BM2F094112; P401
MES_Release	Bool	production release False: production not allowed, don't start machine True: start production	MES		true/false
MES_ProductParameter	Struct	These variables are usually added when additional input parameters from MES to the machine are required.			
MES_CustomerChangeIndex	String[6]	Customer change index ⚠ • The format of the customer change index (from MES field "ordInfoField3") is not fixed. • Get in contact with the DRÄXLMAIER group to get the latest example.	MES		01S
MES_CustomerChangeLevel	String[15]	Customer change level ⚠ • The format of the customer change level (from MES field "ordInfoField2") is not fixed. • Get in contact with the DRÄXLMAIER group to get the latest example.	MES		22.10.2015
MES_CustomerNumber	String[20]	Customer number ⚠ • The format of the customer number (from MES field "ordInfoField1") is not fixed. • Get in contact with the DRÄXLMAIER group to get the latest example.	MES		7641798-03
MES_JobNr	Int16	Can be used to define different product variants that have to be handled on machine side 0: not available (To be defined)	MES		1
MES_PackagingType	String[15]	Box ID (Behälter ID from MES)	MES		APE1
MES_MaterialList	Array[0..x] of Struct	List of needed materials. The format of the list is project specific. Please get in contact with DRX to get a description of the used format. The information has to be checked by the assembly line (valid material numbers, valid box numbers, ...).	MES		
MES_MaterialNr	String[20]	Material number of current material	MES		300000220-001; T0101589
MES_BoxNr	Int16	Box number of current material	MES		2
MES_Handworkplace	Int16	Number of the desired hand-workplace fur current material	MES		5
MES_ToBeDefined		Placeholder for other additional variables. ⚠ • These variables are optional and not mandatory. • These additional variables must be agreed between DRÄXLMAIER and the manufacturer.	MES		e.g. FazitString
Station	Struct	all values written by PLC			
DataAvailable	Bool	False: data not available or not related to current production process True: data available and actual. Must not be set to 0 until `MES_DataProcessed` is set to 1	Station		true/false
Result	Int16	Result of loading process 0: not available 1: OK / INPRODUCTION 2: NOK 3: CANCEL	Station		2
ControlParameter	Struct	Placeholder for needed values which are sent by the machine to the MES			
PartAvailable	Bool	Flag to signalize if a part is inserted in the machine False: no part inserted True: part inserted	Station		true/false
ProductIDAvailable	Bool	False: no part available, or barcode not (yet) scanned automatically True: part available and barcode scanned and filled in ProductID	Station		
ToBeDefined		Placeholder for other additional variables. ⚠ • These variables are optional and not mandatory. • These additional variables must be agreed between DRÄXLMAIER and the manufacturer.	Station		
SecurityData	Struct	Placeholder for security relevant data which should not be visisble in the data structure of processresults. Structure should only be existing if it is in use for this machine/project ⚠ • These variables are optional and not mandatory. • These additional variables must be agreed between DRÄXLMAIER and the manufacturer.			
ProductID	String[100]	Currently processed ProductID (based on MES_ProductID).	Station		Format MES/HOST: BM2F094112CA3320 Format MES2/SAP: 300000220-001P001007827

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CSR	String	CSR String of the product ⚠ The String length of this variable could expand a lot. Maximum String length will be defined within the project.	Station		
PublicKey	String	Public key of the product (base64 encoded DER) ⚠ The String length of this variable could expand a lot. Maximum String length will be defined within the project.	Station		
Signature	String	Signature of the product (base64 encoded) ⚠ The String length of this variable could expand a lot. Maximum String length will be defined within the project.	Station		
ToBeDefined		Placeholder for other additional variables. ⚠ • These variables are optional and not mandatory. • These additional variables must be agreed between DRÄXLMAIER and the manufacturer.	Station		
ProcessResults	Struct	all values written by PLC to specify the process result. The process variables have to be defined together with DRX at the start of the project. Datatypes can be defined by manufacturer according to needed requirements of content.			
GeneralData					
ErrorcodesNOK	Array [0..x] of String[30]	Error codes for each part that was sent as NOK from machine. ⚠ • The error codes have to be defined by the machine manufacturer. • The error codes have to use the format specified in chapter "Format of an Error Number".	Station		[2,0,0,0]
ErrorcodesScrap	Array [0..x] of String[30]	Error codes for each part that was sent as Scrap from machine ⚠ • The error codes have to be defined by the machine manufacturer. • The error codes have to use the format specified in chapter "Format of an Error Number".	Station		[4,0,0,0]
ErrormessagesNOK	Array [0..x] of String[150]	Textual description of an error code from field ErrorcodesNOK ⚠ • One message for each error code in `ErrorcodesNOK` • Errormessage(s) should be collected for each part and have to be deleted after each cycle (MES_DataProcessed=1) • The content may not contain special characters that are also used within an XML notation, e.g. &, <, >	Station		[Torque out of limit,,,]
ErrormessagesScrap	Array [0..x] of String[150]	Textual description of an error code from field ErrorcodesScrap ⚠ • One message for each error code in `ErrorcodesScrap` • Errormessage(s) should be collected for each part and have to be deleted after each cycle (MES_DataProcessed=1) • The content may not contain special characters that are also used within an XML notation, e.g. &, <, >	Station		[Surface too hot,,,]
PartsNOK	Int32	Number of NOK parts for current process. ⚠ • NOK parts can be reworked.	Station		0
PartsOK	Int32	Number of OK parts for current process.	Station		1
PartsScrap	Int32	Number of SCRAP parts for current process.. ⚠ • SCRAP parts can not be reworked.	Station		0
ProductID	String[100]	Currently processed ProductID (based on `MES_ProductID`). ⚠ • See the chapter "ProductID" for a detailled explanation of the used format.	Station		BM2F094126CA3320; 300000220-001P001007827
ProductType	String[50]	Currently processed Product Type (based on `MES_ProductType`).	Station		BM2F094112;300000220-001
AlternativeID	Array [0..x] of String[100]	Barcode (e. g. DMC, Code128, ...) on the final part, that is/are different from the MES internal ProductID and that might be needed for proper tracking of process data in MES. Used to link the MES internal part data to the physical machine part data. ⚠ • The content of the AlternativeID is not fixed and is optional to be filled. • Get in contact with the DRÄXLMAIER group for a final definition if the filling is needed and which value should be filled in.	Station		BUC13092023111299233000022O-001P001007827
TotalProcessingTime	Float	Processing time (duration) for all involved actions in total	Station	minutes	1.87
ProcessTimes	Array of Struct	process times for each process			
EquipmentID	String[30]	ID of the source which reported this time span (e.g. StationID, ToolID…)	Station		ST010_IR01
OperationID	String[20]	Operation which was performed during that time (e.g. pick, driving in, waiting…)	Station		1.0020-10.10
Description	String[50]	short desciption	Station		
StartTime	DateTime (Siemens: LDT)	Start timestamp	Station		18.05.2020 13:00
EndTime	DateTime (Siemens: LDT)	End timestamp	Station		18.05.2020 13:15
MaterialConsumption	Struct	container for material consumption			
AdditionalConsumption_Count	Int16	Amount of actually filled elements in AdditionalConsumption array	Station		5
AdditionalConsumption	Array of Struct	list of all material commissioned additionally (exeeding the regular consumption)			
MaterialNr	String[100]	Host: Material label SAP VDA: Segment P SAP RML: Pos. 1-13	Station		
Consumption	Float	Consumed amount	Station		
Charge	String[30]	Host: Batch label SAP VDA: Supplier or DRÄXLMAIER internal Batch VDA Segment 1T SAP RML: Pos. 14-23	Station		
ExpirationDate	String[20]	optional SAP VDA: Segment 14D SAP RML: n/a	Station		
DeliveryNr	String[30]	optional SAP VDA: Segment 2S SAP RML: n/a	Station		
HandlingUnit	String[20]	SAP VDA: Segment 1J RML: Pos. 24-43	Station		
LabelContent	String[300]	whole 2D barcode label content for VDA including special characters has to be masked, e.g. **#&30;**	Station		
PickPosition	String[20]	optional SAP storage unit where the part was picked (hardcoded on PLC, eg. 0000-000-000-000)	Station		
PositionIndex	String[20]	Detailed position where the material was mounted in the product (optional, only needed if singlepart tracking is required)	Station		
SinglepartLabelSupplier	String[100]	Unique supplier label on the product for traceability. (optional, only needed if singlepart tracking is required)	Station		
RegularConsumption_Count	Int16	Amount of actually filled elements in RegularConsumption array	Station		5
RegularConsumption	Array of Struct	list of all commissioned material			
MaterialNr	String[100]	Host: Material label SAP VDA: Segment P SAP RML: Pos. 1-13	Station		
Consumption	Float	Consumed amount	Station		
Charge	String[30]	Host: Batch label SAP VDA: Supplier or DRÄXLMAIER internal Batch VDA Segment 1T SAP RML: Pos. 14-23	Station		
ExpirationDate	String[20]	optional SAP VDA: Segment 14D SAP RML: n/a	Station		
DeliveryNr	String[30]	optional SAP VDA: Segment 2S SAP RML: n/a	Station		
HandlingUnit	String[20]	SAP VDA: Segment 1J RML: Pos. 24-43	Station		
LabelContent	String[300]	whole 2D barcode label content for VDA including special characters has to be masked, e.g. **#&30;**	Station		
PickPosition	String[20]	optional SAP storage unit where the part was picked (hardcoded on PLC, eg. 0000-000-000-000)	Station		
PositionIndex	String[20]	Detailed position where the material was mounted in the product (optional, only needed if singlepart tracking is required)	Station		
SinglepartLabelSupplier	String[100]	Unique supplier label on the product for traceability. (optional, only needed if singlepart tracking is required)	Station		
UNLOAD		to be used for specific process parameters that are not assigned to a special process (optional)			
FileName	String[50]	If additional data is transferred through Data2Cloud (via a file interface), this variable should contain the respective file name including the extension. To ensure uniqueness the file name should contain the ProductID plus a time stamp. ⚠ • If file transfer is expected, but the machine is unable to save the file in the designated folder, the machine has to return an ErrorMessage to signalize the problem to MES • If no file is to be saved, the value of this field should be empty	Station		A123456701O1_20191205123001.zip
QualityPart	Bool	False: Part was produced normally True: Part was produced for OS reasons	Station		true/false
ToBeDefined		Placeholder for the process parameters of the finished part that don´t belong to a special process	Station		
ToBeDefined		Placeholder for the process parameters of the finished part belonging to the predefined processes for this station. The complete list of process parameters is provided by the DRÄXLMAIER group.	Station		

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